

Frequency Domain Control Design

Ameet Deshpande

September 21, 2026

Abstract

We treat linear control design as a problem in the frequency domain: the plant is known through its frequency responses, the disturbance through its spectrum, and the objective is the real part of the mean inner product of two output vectors, $\text{Re} \mathbb{E}[y_b^* y_a]$ — a mean square when $y_a = y_b$, a mean power when they are a force and a velocity, a cost rate when they are a price and a flow. Nothing is realized in state space. At each frequency the cost is a quadratic form in the controller, which gives a closed-form optimum, a test for when that optimum exists at all, and an expression for the optimal cost that separates what the actuators cannot reach from what the sensors cannot see. Causality, preview, fixed structure and constant gain then appear as one condition on the admissible lags, and the causal optimum follows by Wiener–Hopf factorization, computed on a grid and realized as a finite rational controller by greedy selection of poles. Because sensor noise enters the same quadratic form, the value of sensing falls out of the same algebra: the marginal worth of each sensor’s accuracy, resolved by frequency; a guarantee on what a sensor upgrade is worth, computed before any redesign; and convexity of the accuracy-allocation problem. We give a software package, worked examples on wide-area damping of power systems (a stylized system and the IEEE 39-bus system) and on vibration isolation with two dissimilar sensors. The framework covers one controller with one measurement vector; decentralized problems whose information structure is not quadratically invariant are outside it.

1 Introduction

Two facts about many real control problems sit awkwardly with textbook state-space design. First, what is actually known about the plant is often a set of measured or computed frequency responses — from hydrodynamic codes, aeroelastic codes, system identification, or ambient data — rather than a state-space realization of modest order. Fitting a rational model to that data is a design decision in itself, and a lossy one. Second, what is actually to be optimized is often not a norm. Absorbed power is a force times a velocity; an energy bill is a price times a flow; the energy exchanged during a frequency event is a power times a frequency deviation. Such objectives are bilinear in two different outputs and are not sign-definite, so the standard machinery does not apply unchanged.

This paper works directly with those two facts. The plant enters only through frequency responses, the disturbance only through a spectrum, and the objective is

$$\bar{J} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \text{Re Tr} (Y_a W Y_b^*) d\omega,$$

with y_a and y_b two vectors of objective variables. Because the integrand is a quadratic form in the controller at each frequency, three questions can be answered in closed form and separately at each frequency: what the best controller is, whether a best controller exists at all, and what the resulting cost is. The answers are then constrained back to reality — causality, limited preview, fixed structure — by a single condition, and realized as a controller of finite order.

Four ideas, in plain terms. Four things recur below and are worth stating once without symbols.

1. *Everything is a function of frequency.* Plant, controller and disturbance are described by what they do to a sine wave of each frequency, and the cost is an integral over frequency: a design is judged by how it trades one band against another.
2. *At a single frequency the problem is small.* Fix a frequency and the cost is a quadratic function of one complex matrix, which can be minimized in closed form; the data also say whether a minimum exists at all. Doing this at each frequency separately gives the *pointwise* optimum — a target, usually unachievable, because the answers at different frequencies need not fit together into one realizable controller.
3. *Causality ties the frequencies together.* A causal controller is one whose response to an impulse is zero before the impulse arrives. That is a statement in time, and it constrains all frequencies at once. Preview, constant gain and PID are restrictions of the same kind — on which lags the controller may use — which is why one condition covers them all.
4. *A change of variables makes the problem convex.* The closed loop is a complicated function of the controller but an affine one of the Youla parameter, so designing in that variable is least squares; the controller is recovered afterwards.

What the paper contains. Section 2 gives the pointwise optimum: a stationarity condition, a pseudo-inverse form with the rank condition under which a finite controller realizes it, a test ($H \succeq 0$ at each frequency) that decides whether the problem has a minimum or is unbounded below, and an optimal cost that splits into an actuation-limited part and a sensing gap. Section 3 shows that pointwise, causal, preview, constant-gain and PID designs are all the same stationarity condition restricted to different sets of admissible lags; specializing it to a constant gain under full information returns the LQR condition. Section 4 gives the causal optimum by Wiener–Hopf factorization, for

unstable plants and noisy sensors, with the cost of causality as an explicit quantity into which right-half-plane zeros, delays and unstable poles enter as all-pass factors; it then builds a finite rational controller by greedy pole selection, which is exact after finitely many steps. Section 5 prices sensing. Section 6 describes the software, Section 7 the worked examples, and Section 8 where this should be useful.

What is contributed. Most of the ingredients are established, and Section 9 says which. Stated briefly, this paper adds:

- **A general bilinear objective handled from data.** The pointwise optimum of a power objective is classical for wave-energy arrays [8, 9] and has been carried into a causal design for wave-energy converters [36]. Here the two outputs are arbitrary and so are the sensing and actuation structures; the optimal cost is given in closed form with its sensing gap; the existence test is a pointwise condition on the data rather than a physical passivity assumption; and no rational fit of plant or spectrum is required at any point.
- **One stationarity condition for every structure.** Causality, preview, finite impulse response, constant gain and PID are the same condition on different lag sets (Section 3), which is what makes it possible to compute the cost of causality, the value of preview, and the gap to a fixed-structure design on one scale.
- **The value of sensing, in the frequency domain.** For a fixed controller the cost is affine in the sensor-noise spectrum, so the optimal cost is concave in it and its gradient is an explicit integral (Proposition 8) that says how much each sensor's accuracy is worth and in which frequency band. The same affineness bounds the value of an arbitrary sensor upgrade from the current design alone, before any redesign (Corollary 9), and the allocation of accuracy under a price is a convex problem (Proposition 10).
- **Validated software and examples.** Every frequency-domain result in Section 7 is checked against an exact state-space optimum, including with measurement latency treated exactly as $e^{-s\tau}$, and the code and data are in the repository. The cost of a design depends on the numbers of inputs, sensors and frequency points, not on the plant's state dimension (Section 7.5).

2 Setup and the pointwise optimum

Figure 1 shows the configuration at hand. It shows C the controller and P the plant. The feedback vector is z and u and w are the control and the disturbance inputs, respectively. y_a and y_b are two vectors of objective variables of equal dimension, the real part of whose mean inner product (e.g. mean power, for force and velocity; the mean squared norm, in case the two vectors are the same) we wish to minimize.

Let us denote the corresponding s -domain transfer functions by $C(s)$, $P_{yw}(s)$, $P_{yu}(s)$, $P_{zw}(s)$ and $P_{zu}(s)$. Note that this framework allows for inclusion of the disturbance or

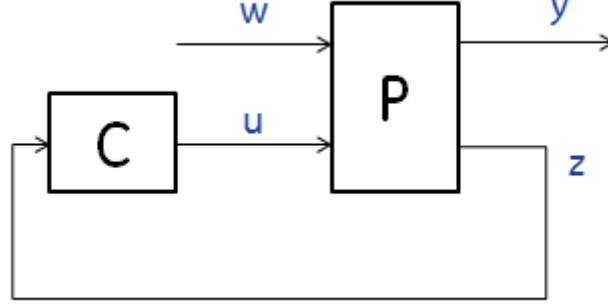


Figure 1: Schematic of the control configuration

reference preview as well as measurement and process noise through P_{zw} . It allows the inclusion of control cost in the objective function through P_{yu} .

All the variables are corresponding complex spectra at a specific frequency, $s = i\omega$. For brevity, this dependence is omitted from the derivation below. The disturbance w is a zero-mean wide-sense-stationary process with spectral density matrix $W(\omega) = W^*(\omega) \succeq 0$; in the per-frequency expressions below, W takes the place of ww^* (a single realization ww^* is only rank one). The notation A^* denotes the complex conjugate transpose of the matrix or vector A , and $A^\dagger$ denotes the Moore-Penrose pseudo-inverse of A , which equals $(A^*A)^{-1}A^*$ only when A has full column rank. The controller is connected in positive feedback, $u = Cz$.

Since we have $u = C(P_{zw}w + P_{zu}u)$, hence

$$u = (I - CP_{zu})^{-1}CP_{zw}w. \quad (1)$$

Using algebra, it is easy to see the following.

$$y_a = P_{y_a u}u + P_{y_a w}w = (P_{y_a u}(I - CP_{zu})^{-1}CP_{zw} + P_{y_a w})w, \quad (2)$$

$$y_b = P_{y_b u}u + P_{y_b w}w = (P_{y_b u}(I - CP_{zu})^{-1}CP_{zw} + P_{y_b w})w. \quad (3)$$

The objective at a specific frequency is the real part of the mean inner product of y_a and y_b . Note that $y_b^*y_a = \text{Tr}(y_a y_b^*)$, whereas $y_a^*y_b$ is its complex conjugate; the two agree only in their real parts, hence

$$\begin{aligned} J(C) &\doteq \text{Re } \mathbb{E}[y_b^*y_a] = \text{Re } \text{Tr}(Y_a W Y_b^*) \\ &= \text{Re } \text{Tr}((P_{y_a u}(I - CP_{zu})^{-1}CP_{zw} + P_{y_a w})W(P_{y_b u}(I - CP_{zu})^{-1}CP_{zw} + P_{y_b w})^*), \end{aligned} \quad (4)$$

where $Y_a \doteq P_{y_a u}(I - CP_{zu})^{-1}CP_{zw} + P_{y_a w}$ and Y_b is defined likewise. The total objective is

$$\bar{J} \doteq \frac{1}{2\pi} \int_{-\infty}^{\infty} J(C(i\omega); \omega) d\omega, \quad (5)$$

and for real systems the integrand is even in ω , so $\bar{J} = \frac{1}{\pi} \int_0^{\infty} J d\omega$.

Taking derivative with respect to the controller C , and equating it to zero, we can solve for the stationary C .

$$\frac{\partial J}{\partial C} = (I_{nz} - \eta(C)) \mathcal{P}(C) \epsilon(C), \quad (6)$$

where

$$\eta(C) \doteq -P_{zu}(I_{nu} - CP_{zu})^{-1}C, \quad (7)$$

$$\begin{aligned} \mathcal{P}(C) \doteq P_{zw}W & \left(P_{zw}^* C^* (I_{nu} - P_{zu}^* C^*)^{-1} (P_{y_a u}^* P_{y_b u} + P_{y_b u}^* P_{y_a u}) \right. \\ & \left. + (P_{y_a w}^* P_{y_b u} + P_{y_b w}^* P_{y_a u}) \right), \end{aligned} \quad (8)$$

$$\epsilon(C) \doteq (I_{nu} - CP_{zu})^{-1}. \quad (9)$$

Here $\epsilon(C)$ is the input sensitivity function, $I_{nz} - \eta(C) = (I_{nz} - P_{zu}C)^{-1}$ is the output sensitivity function, and $\eta(C)$ is the output complementary sensitivity function (these are the standard definitions for the negative-feedback controller $-C$). Also note that $\mathcal{G} \doteq \partial J / \partial C$ in (6) is an $n_z \times n_u$ complex matrix, and the actual change in the real scalar J for a given change δC is, to first order,

$$\delta J = \text{Re Tr} (\mathcal{G} \delta C) = \text{Re} \sum_{i,j} \mathcal{G}_{ji} \delta C_{ij}, \quad (10)$$

so that the direction of steepest descent is $\delta C \propto -\mathcal{G}^*$. For a controller whose response depends on real parameters θ , $\partial J / \partial \theta_k = \text{Re Tr} (\mathcal{G} \partial C / \partial \theta_k)$.

Youla / internal-model form. Define

$$D \doteq C(I_{nz} - P_{zu}C)^{-1} = (I_{nu} - CP_{zu})^{-1}C. \quad (11)$$

By (1), the closed-loop map from w to u is

$$K \doteq (I_{nu} - CP_{zu})^{-1}CP_{zw} = DP_{zw}, \quad (12)$$

which is affine in D . Conversely, the controller is recovered from D as

$$C = (I_{nu} + DP_{zu})^{-1}D \quad (13)$$

whenever $I_{nu} + DP_{zu}$ is invertible. For a stable plant, the closed loop is internally stable if and only if D is stable, so D is the Youla (internal model control) parameter. Substituting (12) into (4),

$$J = \text{Re Tr} ((P_{y_a u}K + P_{y_a w})W(P_{y_b u}K + P_{y_b w})^*). \quad (14)$$

Quadratic form. Define the $n_u \times n_u$ Hermitian matrix H , the $n_u \times n_w$ matrix N , and the open-loop cost J_0 (the cost with $C = 0$):

$$H \doteq P_{y_a u}^* P_{y_b u} + P_{y_b u}^* P_{y_a u}, \quad (15)$$

$$N \doteq P_{y_a u}^* P_{y_b w} + P_{y_b u}^* P_{y_a w}, \quad (16)$$

$$J_0 \doteq \text{Re Tr} (P_{y_a w} W P_{y_b w}^*). \quad (17)$$

Then (14) is exactly the quadratic function (Appendix A)

$$J(K) = \frac{1}{2} \text{Tr} (H K W K^*) + \text{Re Tr} (N^* K W) + J_0. \quad (18)$$

First- and second-order terms. Change K to $K + \delta K$. Because (18) is quadratic, the change in cost is exactly the sum of a term linear in δK and a term quadratic in δK (Appendix A):

$$J(K + \delta K) = J(K) + \underbrace{\text{Re Tr} (G(K) \delta K^*)}_{\text{first-order term}} + \underbrace{\frac{1}{2} \text{Tr} (H \delta K W \delta K^*)}_{\text{second-order term}}, \quad G(K) \doteq (H K + N) W. \quad (19)$$

The first-order term is $\text{Re} \sum_{i,j} G_{ij}(K) \overline{\delta K_{ij}}$, the real inner product of the $n_u \times n_w$ matrix $G(K)$ with δK . It is the only part of the change that matters for small δK , so $G(K)$ is the gradient of J with respect to K , and $\delta K \propto -G(K)$ is the direction of steepest descent. The second-order term does not depend on K ; its sign is decided by H .

The controller can only produce closed-loop maps of the form (12), so the admissible perturbations are $\delta K = \delta D P_{zw}$. With this substitution the first-order term becomes $\text{Re Tr} (G(K) P_{zw}^* \delta D^*)$, which vanishes for every δD if and only if

$$(H K + N) W P_{zw}^* = 0, \quad K = D P_{zw}. \quad (20)$$

This is the same condition as $\mathcal{P}(C) = 0$ in (6), because $\mathcal{P}(C) = [(H D P_{zw} + N) W P_{zw}^*]^*$ (Appendix A).

Stationary controller. Let

$$X_{y_a, y_b} \doteq H^\dagger N, \quad M \doteq P_{zw} W P_{zw}^*, \quad (21)$$

where X_{y_a, y_b} is $n_u \times n_w$ and M is $n_z \times n_z$ Hermitian positive semidefinite. If

$$H H^\dagger N W P_{zw}^* = N W P_{zw}^* \quad (22)$$

(i.e. the columns of $N W P_{zw}^*$ lie in the range of H ; this always holds when H is invertible, and when $y_a = y_b$), then (Appendix B)

$$D_{\text{opt}} = -X_{y_a, y_b} W P_{zw}^* M^\dagger \quad (23)$$

satisfies (20), and C_{opt} follows from (13):

$$C_{\text{opt}} = (I_{nu} + D_{\text{opt}} P_{zu})^{-1} D_{\text{opt}}. \quad (24)$$

Direct expression. The controller can also be written without passing through D . Let

$$Q \doteq (P_{zw} - P_{zu}X_{y_a, y_b})WP_{zw}^*, \quad (25)$$

an $n_z \times n_z$ matrix, and

$$C_{\text{opt}}^d \doteq -X_{y_a, y_b}WP_{zw}^*Q^\dagger. \quad (26)$$

Lemma 1. *Suppose the range condition (22) holds,*

$$\text{rank } Q = \text{rank} \left(P_{zw}W^{1/2} \right), \quad (27)$$

and $I_{n_z} - P_{zu}C_{\text{opt}}^d$ is invertible (the loop is well posed). Then C_{opt}^d satisfies (20), i.e. $\mathcal{P}(C_{\text{opt}}^d) = 0$, and its cost is the same as that of (24). If moreover M is invertible, then $Q^\dagger = Q^{-1}$ and $C_{\text{opt}}^d = C_{\text{opt}}$.

Proof: Appendix B, p. 38.

Some remarks on Lemma 1. M has rank at most n_w , so it is singular whenever $n_z > n_w$. Then (20) has a family of solutions, affine in D but not in C , all with the same cost (Appendix C); (24) and (26) are in general different members of this family. Since $Q = [(P_{zw} - P_{zu}X_{y_a, y_b})W^{1/2}](P_{zw}W^{1/2})^*$, the rank of Q can never exceed $\text{rank}(P_{zw}W^{1/2})$, and (27) holds for generic plant data. When M is invertible, (27) says that Q is invertible, and

$$\det Q = \det M \cdot \det (I_{nu} + D_{\text{opt}}P_{zu}), \quad (28)$$

so (27) fails exactly when no finite controller C realizes D_{opt} through (13); in that case the pointwise optimum is not attainable by any controller of the form $u = Cz$.

Second-order condition.

Proposition 2. *Suppose $M \neq 0$. (a) If $H \succeq 0$, every D satisfying (20) is a global minimizer of J at this frequency. (b) If H has a negative eigenvalue, J is unbounded below at this frequency, and the pointwise problem has no minimizer.*

Proof: Appendix B, p. 38. For $y_a = y_b$, $H = 2P_{yu}^*P_{yu} \succeq 0$ always. For power-type objectives, e.g. minimizing the negative of the generated power, $\text{Re}(u^*v) + u^*Ru$ (mechanical power $-\text{Re}(u^*v)$ minus a quadratic loss), with $v = Zu + \dots$ (take $y_a = [v; u]$, $y_b = [u; Ru]$, $R = R^*$), $H = Z + Z^* + 2R$, which is positive semidefinite when the mapping Z is passive (positive real) and $R \succeq 0$.

Optimal cost. Let $W^{1/2}$ be the Hermitian positive semidefinite square root of W , and define the orthogonal projection onto the disturbance directions that the sensors see,

$$\Pi_z \doteq (P_{zw}W^{1/2})^\dagger(P_{zw}W^{1/2}). \quad (29)$$

Proposition 3. *If (22) holds, every D satisfying (20) gives the same cost,*

$$J_{\text{opt}} = J_0 - \frac{1}{2} \text{Tr} \left(N^* H^\dagger N W^{1/2} \Pi_z W^{1/2} \right). \quad (30)$$

This holds for general $y_a \neq y_b$. When $H \succeq 0$, (30) is the minimum cost (Proposition 2) and can be written as $J_{\text{opt}} = J_0 - \frac{1}{2} \|(H^\dagger)^{1/2} N W^{1/2} \Pi_z\|_F^2 \leq J_0$. When H is indefinite, (30) is the value at a saddle point, not a minimum.

Corollary 4 ($y_a = y_b$). *With the projection $\Pi_u \doteq P_{yu} P_{yu}^\dagger$ onto the output directions the actuators can reach,*

$$J_{\text{opt}} = \text{Tr} \left((I - \Pi_u) P_{yw} W P_{yw}^* \right) + \left\| \Pi_u P_{yw} W^{1/2} (I - \Pi_z) \right\|_F^2. \quad (31)$$

The first term is the part of the disturbance response that the actuators cannot reach. The second term is the part that the actuators could reach but the sensors cannot see; it vanishes when $P_{zw} W^{1/2}$ has full column rank n_w . In the special case of wave-energy absorption — y_a the velocity, y_b the excitation force, and no sensing constraint, so that $\Pi_z = I$ — Propositions 1 and 2 reduce to classical results for interacting oscillators [8], reproduced in [9, §6.4]: the expansion $P(U + \delta) = \frac{1}{4} \hat{F}_e^T U^* - \frac{1}{2} \delta^\dagger R \delta$ about a stationary velocity (their Eq. (6.64)), which is maximal because the radiation-resistance matrix R is positive semidefinite; the maximum absorbed power $\frac{1}{8} \hat{F}_e^\dagger R^{-1} \hat{F}_e$ when R is invertible (their Eq. (6.69)), or $\frac{1}{8} \hat{F}_e^\dagger (R + R_f)^{-1} \hat{F}_e$ with friction losses (their Eq. (6.73)); and, when R is singular, the statement that the optimal velocity is ambiguous while the maximum absorbed power is unique, which is the special case of Proposition 3’s “every stationary D gives the same cost”. Proofs: Appendix C, p. 39 for Proposition 3 and p. 40 for Corollary 4.

Causality. C_{opt} is chosen independently at each frequency, so it is in general non-causal (it anticipates future disturbances). When $H(\omega) \succeq 0$ at all frequencies, $\frac{1}{2\pi} \int J_{\text{opt}} d\omega$ is therefore a lower bound on the total cost (5) of any causal controller, and a causal controller attains it only if some pointwise-optimal D is causal (and stable).

3 Structure-constrained controllers

Condition (20) assumes that $C(i\omega)$ can be chosen independently at every frequency. A realizable controller is usually restricted: a constant gain, a PID controller, a fixed-order transfer function, or any causal controller. This section derives the stationarity condition for such restricted controllers and shows that (20) is the unrestricted special case.

Gradient with respect to C . Write $S_o(C) \doteq (I_{nz} - P_{zu} C)^{-1} = I_{nz} - \eta(C)$ for the output sensitivity. Perturbing the controller from C to $C + \delta C$ changes the Youla parameter (11) by (Appendix D)

$$\delta D = \epsilon(C) \delta C S_o(C) + O(\|\delta C\|^2), \quad (32)$$

so the closed-loop map changes by $\delta K = \epsilon \delta C S_o P_{zw}$ to first order. Substituting into the first-order term of (19),

$$\delta J = \text{Re Tr} (\Gamma(C; \omega) \delta C^*) + O(\|\delta C\|^2), \quad \Gamma(C; \omega) \doteq \epsilon^* (HK + N) W P_{zw}^* S_o^*, \quad (33)$$

where Γ is $n_u \times n_z$ and $K = DP_{zw}$ is the closed-loop map of C . Γ is the conjugate transpose of $\mathcal{G}$ in (6) and (10). Since ϵ and S_o are invertible,

$$\Gamma(C; \omega) = 0 \iff (HK + N) W P_{zw}^* = 0, \quad (34)$$

which is (20).

Stationarity for a restricted controller. Define the real inner product of two $n_u \times n_z$ matrix functions of frequency,

$$\langle X, Y \rangle \doteq \frac{1}{2\pi} \int_{-\infty}^{\infty} \text{Re Tr} (X(\omega) Y(\omega)^*) d\omega. \quad (35)$$

By (5) and (33), the total cost changes by $\delta \bar{J} = \langle \Gamma(C), \delta C \rangle$ to first order. Let the admissible controllers be $C(i\omega) = C(i\omega; \theta)$ with real parameters $\theta \in \mathbb{R}^p$. A stationary point satisfies

$$\begin{aligned} \frac{\partial \bar{J}}{\partial \theta_k} &= \left\langle \Gamma(C), \frac{\partial C}{\partial \theta_k} \right\rangle = \frac{1}{2\pi} \int_{-\infty}^{\infty} \text{Re Tr} \left(\Gamma(C; \omega) \left(\frac{\partial C(i\omega; \theta)}{\partial \theta_k} \right)^* \right) d\omega \\ &= 0, \quad k = 1, \dots, p. \end{aligned} \quad (36)$$

In words, Γ must be orthogonal to every admissible direction of change of the controller. If $C(i\omega)$ may be chosen freely at each frequency, the admissible directions include every matrix function of ω , and (36) forces $\Gamma \equiv 0$, i.e. (34) and hence (20). Condition (36) is therefore the generalization of (20).

Time-domain form. For real systems, real controllers, and a real disturbance process, $\Gamma(C; -\omega) = \overline{\Gamma(C; \omega)}$, so

$$\gamma(t) \doteq \frac{1}{2\pi} \int_{-\infty}^{\infty} \Gamma(C; \omega) e^{i\omega t} d\omega \quad (37)$$

is a real $n_u \times n_z$ matrix function of the lag t . If the admissible controllers are exactly those whose impulse response $c(t)$ may be chosen freely for t in a set $\mathcal{S}$ of lags (and is zero elsewhere), Parseval's theorem turns (36) into (Appendix D)

$$\gamma(t) = 0 \quad \text{for all } t \in \mathcal{S}. \quad (38)$$

This gives a hierarchy of conditions:

- unrestricted, non-causal controller, $\mathcal{S} = \mathbb{R}$: $\gamma \equiv 0$, which is (20);
- causal controller, $\mathcal{S} = [0, \infty)$: $\gamma(t) = 0$ for $t \geq 0$, a Wiener-Hopf condition, with $\gamma(t)$ free for $t < 0$;

- controller with preview τ and memory T , $\mathcal{S} = [-\tau, T]$: $\gamma(t) = 0$ on that interval;
- constant real gain, $c(t) = C \delta(t)$, $\mathcal{S} = \{0\}$:

$$\gamma(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \Gamma(C; \omega) d\omega = 0. \quad (39)$$

For a PID controller $C(s) = K_P + K_I/s + K_D s$, with real matrices K_P, K_I, K_D , (36) gives three matrix conditions, assuming the integrals converge (in practice the derivative term is filtered):

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \Gamma(C; \omega) \overline{b(i\omega)} d\omega = 0, \quad b(s) \in \left\{ 1, \frac{1}{s}, s \right\}. \quad (40)$$

The first is (39). Formally, the second says that $\int_0^\infty \gamma(t) dt = 0$ (the impulse response of $1/s$ is a unit step).

Full-state feedback and LQR. Now impose the constant-gain constraint on the full-state problem: $z = x$ is measured without noise, so $P_{zw} = \Phi B_w$ and $P_{zu} = \Phi B_u$ with $\Phi(i\omega) = (i\omega I - A)^{-1}$; the disturbance is white with $W = I$; and $y_a = y_b = [Q^{1/2}x; R^{1/2}u]$ with $Q \succeq 0$, $R \succ 0$. For a real constant gain C with $A_c \doteq A + B_u C$ Hurwitz, let Σ_C and P_C solve the Lyapunov equations

$$A_c \Sigma_C + \Sigma_C A_c^T + B_w B_w^T = 0, \quad A_c^T P_C + P_C A_c + Q + C^T R C = 0. \quad (41)$$

Proposition 5. *Under these assumptions:*

- (a) *the frequency-domain constant-gain gradient (39) equals the Levine-Athans gradient,*

$$\gamma(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \Gamma(C; \omega) d\omega = 2 (B_u^T P_C + R C) \Sigma_C; \quad (42)$$

- (b) *if $\Sigma_C \succ 0$, then $\gamma(0) = 0$ if and only if $C = -R^{-1} B_u^T P$, where P is the stabilizing solution of the algebraic Riccati equation $A^T P + P A - P B_u R^{-1} B_u^T P + Q = 0$. That is, the only stationary stabilizing constant gain is the LQR gain.*

Proof: Appendix D, p. 41. Two caveats on attribution. First, (42) is the gradient used throughout the static output-feedback literature and attributed to [23], whose necessary condition for a constrained gain $u = -FCx$ is usually quoted in the coupled form $F = R^{-1} B_u^T K L C^T (C L C^T)^{-1}$ with K a Riccati-like matrix and L a state covariance; its full-information case is the LQR gain, so part (b) is a specialization of that condition and not a new result. We have not obtained [23]; what is verified here is the formula, not the attribution: `src/verify_structured.py` evaluates the frequency-domain gradient (39) and $2(B_u^T P_C + R C) \Sigma_C$ at several gains and they agree to six decimals, and gradient descent using only the frequency-domain quantity converges to $-R^{-1} B_u^T P$. Second, in [23] the matrix Σ_C arises from averaging over random initial states rather than from a

white disturbance; the two give the same gradient with $B_w B_w^T$ replaced by the corresponding second-moment matrix. What is contributed here is the route: (42) is what the lag-set stationarity condition (38) becomes when the admissible set is a single lag at $t = 0$.

So imposing the constant-gain constraint on the frequency-domain stationarity condition recovers LQR. In the example of Figure 2 ($A = [0, 1; -10, -1]$, $B_u = [0; 1]$, $B_w = [1; 0.3]$, $Q = I$, $R = 0.1$), the following checks agree:

- (42) matches the Lyapunov form to six digits at four different gains;
- a quasi-Newton descent that uses only (39) as its gradient, starting from $C = 0$, converges to $-K_{\text{LQR}} = [-0.4881, -2.4607]$ with $\bar{J} = 2.681$.

Solving the condition, and what it does not give. (42) is a gradient, not a formula for the gain. For a general measured output the classical procedure iterates the two Lyapunov equations (41) and a gain update read off the same condition, restarting from a stabilizing gain [23]; the descent above does the same work from (39) alone, which needs only frequency responses and no state-space model. Either way the condition is necessary, not sufficient, and the constraint is not of the kind that preserves the convexity of Section 1: causality, preview and finite-lag constraints are linear in the Youla parameter, whereas holding the *controller* constant is not, so the cost need not be convex in the gain. The landscapes of static output feedback and of LQG have been studied for exactly this reason [10, 45], and several stationary points are possible.

The cost does not enforce stability. A second warning matters more in practice. The cost (5) is an integral over the imaginary axis, and it equals $\mathbb{E}[y_b^* y_a]$ only when the closed loop is stable; off the stabilizing set the same formula returns a finite number that means nothing. One might hope that descent could not leave the stabilizing set, because a closed-loop pole crossing the imaginary axis at $\omega_0 \neq 0$ makes $|S(i\omega_0)|$ infinite and the integral diverge. That barrier is real, but it can be absent. A pole can also leave through the origin: for a controller with integral action, $\mathcal{I}$ the integral gain, this happens when $\det(P_{zu}(0)\mathcal{I})$ changes sign, and there is no barrier at all, because $S(0) = 0$ on both sides and the transit contributes at a single frequency. In the PID example of Section 7 this is exactly what happens: along the segment from a stabilizing design to the minimizer of the spectral cost, $\det(P_{zu}(0)\mathcal{I})$ passes through zero, the number of unstable closed-loop poles goes from zero to two, and the cost rises only from 0.018 to 0.033 before falling to 0.003 — a bump that a line search steps over. With the Youla parameterization of Section 4 the question does not arise, since a stable Q gives internal stability by construction; when optimizing a structured controller directly, a stability certificate has to be carried alongside the cost. Proposition 5(b) is the benign case: with full information and $\Sigma_C \succ 0$ the stationary stabilizing gain is unique. We make no claim about the computational complexity of the constrained problem.

The LQR gain satisfies more than (39). Because LQR is optimal among all causal controllers for this problem, $\gamma(t) = 0$ for every $t \geq 0$ (numerically, $\max_{t \geq 0.05} |\gamma(t)| \approx$

10^{-7}), while $\gamma(t) \neq 0$ for $t < 0$ (Figure 2). The non-zero part at $t < 0$ is exactly what (20) would also remove. It is the improvement available only to a non-causal controller: here the pointwise optimum has cost 1.69 against 2.68 for LQR.

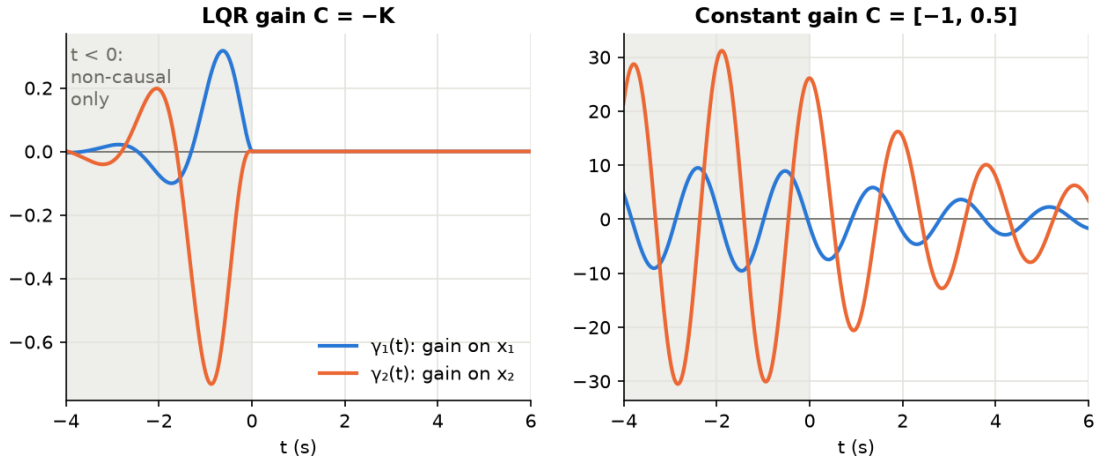


Figure 2: The stationarity kernel $\gamma(t)$ of (37) for full-state feedback with white noise. Left: at the LQR gain, $\gamma(t) = 0$ for all $t \geq 0$, so the gain is stationary among constant gains ($\gamma(0) = 0$) and among all causal controllers, but not in the pointwise sense (20). Right: a non-optimal constant gain, for which $\gamma(0) \neq 0$.

For output feedback (for example static output feedback or PID), (36) is still the first-order condition, but a stationary point need not be a global minimum: the cost is not convex in the controller parameters, and suboptimal stationary points can exist (see Fatkhullin and Polyak, and Zheng, Tang and Li, on the optimization landscape of static output feedback and of LQG).

4 Causal optimum, unstable plants, and sensor noise

Section 3 gives the causal stationarity condition $\gamma(t) = 0$ for $t \geq 0$. This section solves it in closed form. The solution also covers unstable plants, plants with right-half-plane (RHP) zeros or delays, and noisy sensors.

Youla parameterization for a general plant. For an unstable plant, a stable D no longer guarantees internal stability. Using a coprime factorization of P_{zu} , the closed-loop maps of all internally stabilizing controllers can be written as affine functions of a free stable parameter (Boyd and Barratt, Eq. (7.25); Skogestad and Postlethwaite, Eq. (4.94)). In particular, the map from w to u is

$$K = K_0 + T_2 Q T_3, \quad Q \text{ stable and causal}, \quad (43)$$

where K_0 is the closed-loop map of one stabilizing controller C_0 , and T_2, T_3 are stable. A stable plant is the special case $K_0 = 0$, $T_2 = I$, $T_3 = P_{zw}$, $Q = D$. For a single-loop

plant $P_{zu} = N_c/M_c$, with M_c stable and all-pass and carrying the unstable poles, one can take $K_0 = D_0 P_{zw}$ with D_0 the Youla parameter (11) of C_0 , $T_2 = M_c$, and $T_3 = M_c P_{zw}$. Substituting (43) into (19) gives the same quadratic structure as before (Appendix E):

$$J = J(K_0) + \text{Re Tr}(E Q^*) + \frac{1}{2} \text{Tr}(\tilde{H} Q \tilde{M} Q^*), \quad (44)$$

$$\tilde{H} \doteq T_2^* H T_2, \quad \tilde{M} \doteq T_3 W T_3^*, \quad E \doteq T_2^* (H K_0 + N) W T_3^*. \quad (45)$$

For a stable plant, $\tilde{H} = H$, $\tilde{M} = M$ and $E = N W P_{zw}^*$.

Sensor noise. If the sensors are noisy, $z = P_{zw}w + P_{zu}u + v$, where v has spectral density $V \succeq 0$ and is uncorrelated with w , then no new theory is needed. Augment the disturbance to $[w; v]$:

$$P_{zw} \rightarrow [P_{zw} \ I], \quad P_{y_a w} \rightarrow [P_{y_a w} \ 0], \quad P_{y_b w} \rightarrow [P_{y_b w} \ 0], \quad W \rightarrow \begin{bmatrix} W & 0 \\ 0 & V \end{bmatrix}. \quad (46)$$

Then $M = P_{zw} W P_{zw}^* + V$, which is positive definite when $V \succ 0$. Adding sensors can never increase the optimal cost, pointwise or causal, because the controller may give a new sensor zero weight. Without noise, however, $n_z > n_w$ sensors are redundant: M is singular, the optimal controller is not unique (Lemma 1), and the causal formula below cannot be applied directly. In some problems the noise-free optimum even requires differentiating the measurements, i.e. an improper controller. Modelling the (always present) sensor noise with $V \succ 0$ removes this degeneracy. This is the frequency-domain counterpart of the requirement of positive-definite measurement noise in a Kalman filter.

Causal optimum. For matrix functions of frequency, write $\|X\|^2 \doteq \frac{1}{2\pi} \int_{-\infty}^{\infty} \|X(\omega)\|_F^2 d\omega$. Write $[X]_+$ and $[X]_-$ for the causal and anticausal parts of X : the Fourier transforms of the $t \geq 0$ and $t < 0$ halves of its inverse transform (37), so that $X = [X]_+ + [X]_-$ and $\|X\|^2 = \|[X]_+\|^2 + \|[X]_-\|^2$. That is: transform to time, keep what depends on the past, discard what would depend on the future. What is discarded is what a causal controller cannot have, and its size is the price of causality.

Theorem 6. Assume that $\tilde{H}(\omega) \succeq \epsilon I$ and $\tilde{M}(\omega) \succeq \epsilon I$ for some $\epsilon > 0$ and all ω , including $\omega \rightarrow \infty$. Assume also that they have spectral factorizations

$$\tilde{H} = \Lambda^* \Lambda, \quad \tilde{M} = \Gamma \Gamma^*, \quad (47)$$

with $\Lambda, \Lambda^{-1}, \Gamma, \Gamma^{-1}$ stable and causal (for real-rational data these exist under the stated positivity; see the matrix spectral factorization of Youla (1961), as referenced by Boyd and Barratt). Such a factorization is a square root of a positive spectrum that can be built as a causal filter and undone by one, the role a Cholesky factor plays for a positive matrix. Let

$$F \doteq \Lambda^{-*} E \Gamma^{-*}, \quad (48)$$

and assume $\|F\| < \infty$. Then among stable, causal Q (with $\Lambda Q \Gamma$ square integrable), the optimum and its cost are

$$Q_c = -\Lambda^{-1}[F]_+ \Gamma^{-1}, \quad (49)$$

$$\bar{J}(Q_c) = \bar{J}_{\text{pw}} + \frac{1}{2} \|[F]_-\|^2, \quad (50)$$

where the total cost of the pointwise optimum is $\bar{J}_{\text{pw}} = \bar{J}(K_0) - \frac{1}{2} \|F\|^2$, attained by $Q_{\text{pw}} = -\Lambda^{-1} F \Gamma^{-1}$.

Proof: Appendix E, p. 42.

So the price of causality is $\frac{1}{2} \|[F]_-\|^2$, the energy of the anticausal part of F . For a stable plant with invertible H and M , $Q_{\text{pw}} = -H^{-1} N W P_{zw}^* M^{-1} = D_{\text{opt}}$ of (23). The requirement $\tilde{H} \succeq \epsilon I$ as $\omega \rightarrow \infty$ needs a control-effort term in y_a, y_b , as LQR needs $R \succ 0$; $\tilde{M} \succeq \epsilon I$ needs sensor noise. For an indefinite $\tilde{H}$ (power-type objectives), the cost is unbounded below by Proposition 2, and no causal optimum exists either.

RHP zeros, delays, and unstable poles. These enter F as all-pass factors: magnitude one at every frequency, so they change only phase and no stable causal filter can undo them. Suppose (single input) $P_{yu} = U P_0$ with U all-pass ($U^* U = 1$ on the imaginary axis, containing the RHP zeros or a delay e^{-sT}) and P_0 minimum phase. Then $H = 2P_0^* P_0$, so the spectral factor Λ does not see U . But N contains $P_{yu}^* = P_0^* U^*$, so F contains U^* , which is anticausal: for a delay, $U^* = e^{sT}$ shifts the impulse response of F earlier by T . This moves energy into $[F]_-$ and raises the causality cost (50). For a delay, a preview of T seconds is exactly what removes it. The all-pass factor M_c of an unstable plant enters E through $T_2^* = M_c^*$ and T_3^* in the same way.

Numerical check. Table 1 compares the Wiener-Hopf solution (49), computed on a frequency grid ($dt = 0.005$ s, band ± 628 rad/s) with cepstral scalar spectral factors, against LQG computed from Riccati equations. The plant is the example of Figure 2: $A = [0, 1; -10, -1]$, $B_u = [0; 1]$, $B_w = [1; 0.3]$, $y = [x; R^{1/2}u]$, $R = 0.1$, with sensors $z = x + v$ (two sensors) or $z = x_1 + v$ (one sensor), $V = \sigma^2 I$. For two sensors, Γ is taken from the Kalman filter, $\Gamma = (I + \Phi L) V^{1/2}$, which factors M to 5×10^{-14} .

The Wiener-Hopf and LQG costs agree within 0.05–0.4%. The remaining difference is a discretization effect, not a modelling one: on a sampled, band-limited grid the causal projection $[\cdot]_+$ is exact only as $dt \rightarrow 0$. For the RHP zero at $a = 3$, the difference to LQG evaluated on the same grid is 1.1×10^{-2} , 5.6×10^{-3} and 2.9×10^{-3} for $dt = 0.02$, 0.01 and 0.005 s: it halves with dt . Some observations:

- The pointwise optimum does not depend on the RHP zero at all (column $\bar{J}_{\text{pw}}$): an all-pass factor changes only the phase, which the pointwise optimum can undo at every frequency. The causality cost grows most when the zero is near the frequencies that matter (here the 3.2 rad/s resonance); a much slower zero ($a = 0.3$) hurts less.

Plant	Sensors, σ	$\bar{J}_{\text{pw}}$	$\bar{J}$ Wiener-Hopf (49)	$\bar{J}$ LQG	causality cost
stable	$x_1, x_2; 1.0$	2.340	3.746	3.749	1.406
stable	$x_1; 1.0$	4.090	4.931	4.932	0.841
stable	$x_1, x_2; 0.1$	1.707	2.717	2.718	1.010
stable	$x_1; 0.1$	1.793	2.868	2.868	1.074
stable	$x_1, x_2; 0.03$	1.697	2.686	2.687	0.989
stable	$x_1; 0.03$	1.705	2.702	2.703	0.997
RHP zero $a = 30$	$x_1; 0.1$	1.793	2.875	2.876	1.082
RHP zero $a = 10$	$x_1; 0.1$	1.793	2.965	2.968	1.172
RHP zero $a = 3$	$x_1; 0.1$	1.793	3.605	3.611	1.812
RHP zero $a = 1$	$x_1; 0.1$	1.793	3.653	3.656	1.860
RHP zero $a = 0.3$	$x_1; 0.1$	1.793	3.176	3.177	1.383
unstable (pole +1)	$x_1; 0.1$	0.560	2.929	2.939	2.369

Table 1: Causal optimum from Theorem 6 versus LQG. The causality cost is $\frac{1}{2}\|[F]_-\|^2 = \bar{J}_{\text{WH}} - \bar{J}_{\text{pw}}$. RHP zero: the actuator passes through the all-pass $(a-s)/(a+s)$. Unstable: $A = [0, 1; 2, -1]$ (poles +1, -2), with K_0 from a deliberately non-optimal stabilizing controller C_0 (cost 2.964) and $M_c = (s-1)/(s+1)$.

- Two sensors beat one at every noise level, and as $\sigma \rightarrow 0$ both approach the full-state LQR cost 2.681.
- For the unstable plant, the causal optimum improves on the stabilizing C_0 and matches LQG, and the gap to the pointwise bound is large (2.37): stabilization is costly, and the pointwise bound ignores it.

Finite causal bases. In practice, Q (here D , for a stable plant) is restricted to a finite causal basis,

$$D(s) = \sum_{k=1}^n \theta_k B_k(s), \quad \theta_k \in \mathbb{R}, \quad (51)$$

with each B_k stable and causal. Then every such D gives a stable closed loop, and by (44) the cost is a quadratic in θ . For a single input and single sensor it is

$$\bar{J}(\theta) = \bar{J}(K_0) + b^T \theta + \frac{1}{2} \theta^T \mathbf{H} \theta, \quad \mathbf{H}_{kl} = \langle \tilde{H} \tilde{M} B_k, B_l \rangle, \quad b_k = \langle E, B_k \rangle, \quad (52)$$

with $\langle \cdot, \cdot \rangle$ from (35). Its minimizer solves the linear equations $\mathbf{H} \theta = -b$, which is convex whenever $\tilde{H} \succeq 0$. Two standard bases of stable, causal functions, orthonormal in $\|\cdot\|$,

are

$$\text{Laguerre: } B_k(s) = \frac{\sqrt{2a}}{s+a} \left(\frac{s-a}{s+a} \right)^{k-1}, \quad (53)$$

$$\begin{aligned} \text{Kautz: } B_{2k-1}(s) &= \frac{\sqrt{2\beta}s}{s^2 + \beta s + \kappa} \Psi^{k-1}, \quad B_{2k}(s) = \frac{\sqrt{2\beta\kappa}}{s^2 + \beta s + \kappa} \Psi^{k-1}, \\ \Psi &\doteq \frac{s^2 - \beta s + \kappa}{s^2 + \beta s + \kappa}, \end{aligned} \quad (54)$$

with a real pole $-a$ for Laguerre and a pole pair $s^2 + \beta s + \kappa$ for Kautz.

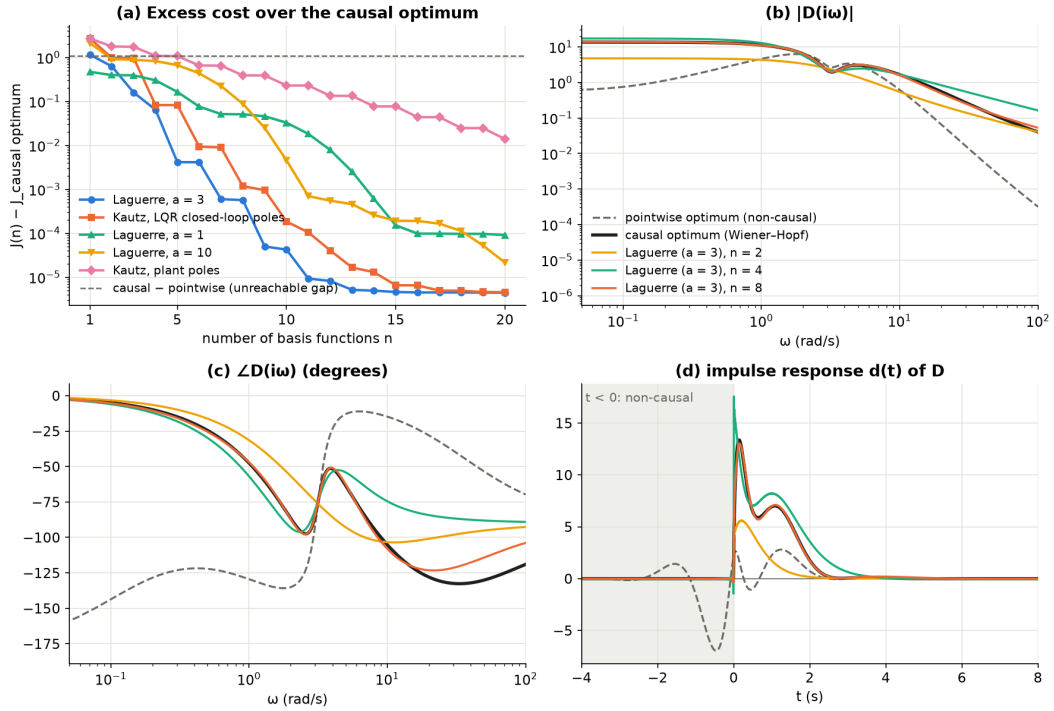


Figure 3: Restricting the Youla parameter D to n causal basis functions (stable plant, one sensor, $\sigma = 0.1$). (a) Excess cost over the causal optimum; the dashed line is the (unreachable) gap between the causal and pointwise optima, and the floor near 5×10^{-6} is the grid discretization error. (b, c) Magnitude and phase of $D(i\omega)$ for Laguerre ($a = 3$) with $n = 2, 4, 8$, the causal optimum (49), and the non-causal pointwise optimum (23). (d) The corresponding impulse responses: the basis fits and the causal optimum vanish for $t < 0$, while the pointwise optimum does not.

Figure 3 shows the one-sensor example ($\sigma = 0.1$). As n grows, the cost converges to the causal optimum 2.868 and not to the pointwise value 1.793. The fitted $D(i\omega)$ likewise converges to the causal optimum, which differs substantially from the non-causal pointwise optimum, especially below the resonance. The speed of convergence depends on where the basis poles are:

Basis	$n = 2$	$n = 4$	$n = 8$	$n = 20$
Laguerre, $a = 3$	3.494	2.932	2.868	2.868
Kautz, poles at the LQR closed-loop poles $-1.73 \pm 2.74i$	3.848	2.950	2.869	2.868
Laguerre, $a = 1$	3.264	3.170	2.918	2.868
Laguerre, $a = 10$	3.782	3.694	2.956	2.868
Kautz, poles at the plant poles $-0.5 \pm 3.12i$	4.640	3.954	3.259	2.882

For a stable plant, $D = C(I - P_{zu}C)^{-1}$ is itself a closed-loop map, so its poles are closed-loop poles, not plant poles. A basis should therefore place its poles near the expected closed-loop poles (roughly, near the closed-loop bandwidth). A Laguerre basis with a single real pole at that bandwidth is a good default because it has only one parameter to choose. A Kautz or other rational basis does as well when its poles are placed at (estimates of) the closed-loop poles, but poorly when placed at lightly damped plant poles. In practice one can fit with Laguerre, read off the dominant poles of the result, and refit with a rational basis at those poles.

Learning the basis pole. The Laguerre pole a need not be known in advance. For fixed a , the optimal coefficients solve the linear equations of (52); call the resulting cost $\bar{J}^*(a) = \min_{\theta} \bar{J}(\theta, a)$. This is a function of a single real number. Because the coefficients are already optimal for the current pole, moving them changes nothing to first order, so the pole may be differentiated with the coefficients held fixed (the envelope theorem):

$$\frac{d\bar{J}^*}{da} = \frac{1}{2} \theta^{*T} \frac{\partial \mathbf{H}}{\partial a} \theta^* + \left(\frac{\partial b}{\partial a} \right)^T \theta^*, \quad \frac{\partial B_k}{\partial a} = B_k \left(\frac{1}{2a} - \frac{k-1}{s-a} - \frac{k}{s+a} \right). \quad (55)$$

Figure 4 shows what happens. $\bar{J}^*(a)$ has local minima and is nearly flat for small a (panel a), so plain gradient descent on $\log a$ depends on its starting point: for $n = 8$, starting at $a = 0.1$ stays at $a = 0.1$ ($\bar{J} = 3.259$), while starting at $a = 30$ reaches $\bar{J} = 2.868$ (panel b). Since the search is one-dimensional, a cheap remedy works without any knowledge of the closed loop: evaluate $\bar{J}^*$ at 16 log-spaced values $a \in [10^{-2}, 10^3]$ rad/s, refine the three best by gradient descent, and keep the best. The learned pole does as well as the best hand-picked pole at every n (panel c) and settles at $a^* \approx 2\text{--}4.5$ rad/s (panel d), close to the magnitude 3.24 of the LQR closed-loop poles, which the method never uses. Near the optimum, $\bar{J}^*(a)$ is rugged at the 10^{-4} level, so the learned a^* varies with n ; for $n = 8$ a dense 60-point sweep finds a minimum only 1.3×10^{-4} lower. With several poles (a generalized orthonormal basis), the same gradient (55) applies pole by pole, but the search is no longer one-dimensional and needs more care.

Orthogonality and greedy pole selection. Laguerre (and Kautz) functions are orthonormal in the plain norm $\|\cdot\|$, but the cost (52) uses the weighted Gram matrix $\mathbf{H}$, so the coefficients are coupled and adding a function changes all earlier coefficients. Orthogonality becomes useful in the whitened coordinates of Theorem 6. Write $Y \doteq \Lambda Q \Gamma$, so that, by (94), $\bar{J} = \text{const} + \frac{1}{2} \|Y + F\|^2$. Expanding $Y = \sum_k \theta_k \phi_k$ in any basis of stable,

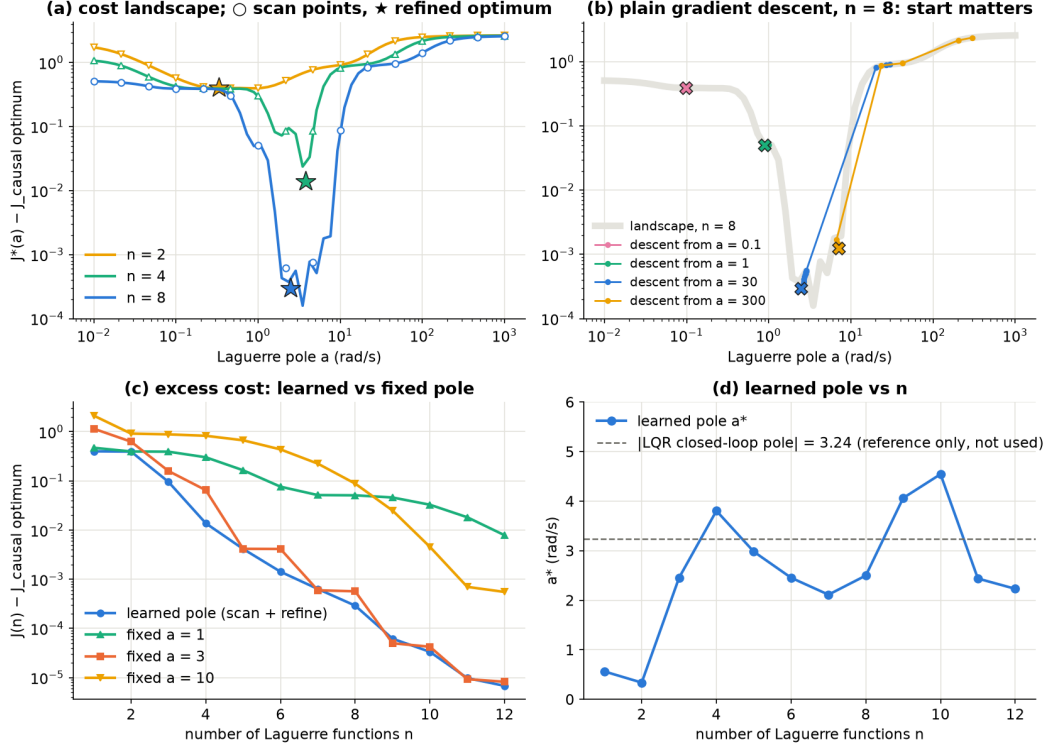


Figure 4: Learning the Laguerre pole (same example as Figure 3). (a) $\bar{J}^*(a)$ minus the causal optimum for $n = 2, 4, 8$; circles are the 16 scan points, stars the refined optima. (b) Plain gradient descent on $\log a$ for $n = 8$ from four starting points. (c) Excess cost against n for the learned pole and for fixed poles $a = 1, 3, 10$. (d) The learned pole against n ; the dashed line is the magnitude of the LQR closed-loop poles, shown for reference only.

causal functions that is orthonormal in $\|\cdot\|$ gives

$$\theta_k = -\langle F, \phi_k \rangle = -\langle [F]_+, \phi_k \rangle, \quad \bar{J}_n - \bar{J}(Q_c) = \frac{1}{2} \sum_{k>n} \theta_k^2, \quad (56)$$

where $\bar{J}_n$ is the cost with the first n functions. Each coefficient is a single inner product, adding a function never changes the earlier ones, and the remaining excess cost is known exactly. How fast the series converges depends on the poles of the basis relative to those of the target $[F]_+$; for generalized orthonormal bases this is quantified by Heuberger, Van den Hof and Bosgra (1995, Theorem 8.3): the coefficients decay geometrically at a rate set by the distance between basis poles and target poles, and they vanish after finitely many terms when the poles coincide. In the whitened coordinates the relevant poles are known in principle:

Proposition 7. *For real-rational data satisfying the assumptions of Theorem 6, $[F]_+$ is*

rational and its poles are among the left-half-plane poles of E . Hence the causal optimum $Y_c = -[F]_+$ is a finite combination of Takenaka-Malmquist functions with those poles.

Proof: Appendix F, p. 42.

For a stable plant with a white disturbance, $E = NW P_{zw}^*$ and the left-half-plane poles of E are the poles of P_{yaw}, P_{ybw} , i.e. plant (and disturbance-model) poles. This explains the observation above that Kautz functions at the plant poles converge poorly for D (whose poles are closed-loop poles) but are exactly the right basis for Y .

When the poles are not known, they can be found greedily. At step $n + 1$, form for a candidate pole p (real, or a conjugate pair) the real-coefficient functions $\psi(s) \in \{\frac{s}{(s-p)(s-\bar{p})}, \frac{1}{(s-p)(s-\bar{p})}\}$ (or $\frac{1}{s-p}$) multiplied by the Blaschke product $\prod_{j \leq n} \frac{(s+p_j)(s+\bar{p}_j)}{(s-p_j)(s-\bar{p}_j)}$ of the poles already chosen, which makes them orthogonal to all earlier functions. Choose the pole whose functions capture the most of the current residual $R_n = [F]_+ + Y_n$:

$$p_{n+1} = \arg \max_p \|\Pi_p R_n\|^2, \quad (57)$$

where Π_p is the orthogonal projection onto the span of the candidate functions. This is the maximal selection principle of adaptive Fourier decomposition (Qian and Wang; see Gao, Ku and Qian, Eqs. (5)–(8)), here in the right half-plane with real coefficients and applied to the causal target of Theorem 6. Without spectral factors (e.g. for several inputs, where matrix factorization is harder), the same greedy selection can be done directly on D (or Q) in the weighted metric of (52): for a candidate block of new columns, the exact cost decrease at the current optimum θ^* is (Appendix F)

$$\Delta \bar{J} = \frac{1}{2} \rho^T S^{-1} \rho, \quad \rho = b_\psi + \mathbf{H}_\psi \theta^*, \quad S = \mathbf{H}_\psi \psi - \mathbf{H}_\psi \theta \mathbf{H}_\theta^{-1} \mathbf{H}_\theta \psi, \quad (58)$$

where ρ is the gradient with respect to the new coefficients and S is a Schur complement. Candidates are evaluated on a logarithmic grid of poles and the best one is refined by a local (Nelder-Mead) search.

Figure 5 compares five choices on the example of Figures 3–4. Greedy selection on Y picks the plant pole pair $-0.50 \pm 3.12i$ first and reaches the grid-error floor with two functions, as Proposition 7 predicts. Laguerre functions on Y converge slowly (the target is resonant). On D , the one-step-ahead greedy rule is myopic and, in this example, slower than a Laguerre basis with a learned pole; its advantage is that it needs no pole guess and extends unchanged to several inputs and outputs and to unstable plants.

Discretization artifacts. On a finite frequency grid, a basis function whose pole lies near the edge of the band is not fully resolved, and a greedy search can then lower the cost *as measured on the grid* by moving response above the band, where it is not counted. Two safeguards remove this: poles are restricted to a band well inside the grid ($|\operatorname{Re} p|, \operatorname{Im} p \leq 0.1\pi/dt$), and each step's controller, being an explicit rational function, is re-evaluated on a finer validation grid; tuning stops when the validated cost stops decreasing. Without these safeguards the greedy search reported costs below the exact causal optimum in our tests; with them, the validated costs agree with the exact optimum (Table 2).

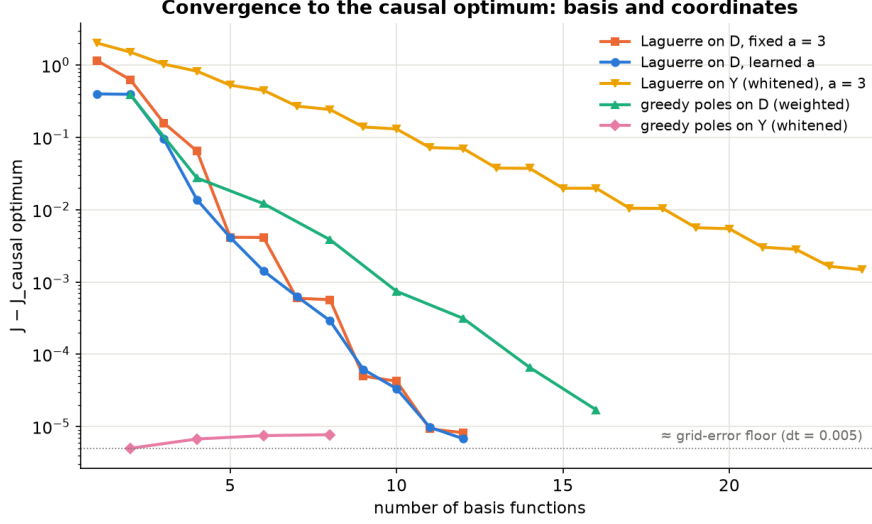


Figure 5: Excess cost over the causal optimum against the number of basis functions (one sensor, $\sigma = 0.1$, white disturbance). Laguerre on D with fixed or learned pole; Laguerre on the whitened $Y = \Lambda D \Gamma$; greedy pole selection on D in the weighted metric, (58), and on Y , (57).

5 The value of sensing

Sensor noise enters the cost through the same quadratic form as everything else, so what a sensor is worth can be read off the design that is already in hand. This section makes that precise: how much each sensor's accuracy is worth and at which frequencies, what an upgrade is guaranteed to buy, and why allocating accuracy under a price is a tractable problem.

Sensor noise enters only through the augmented spectrum (46). For a stable plant with the parameterization $Q = D$, the noise columns of $P_{y_a w}$, $P_{y_b w}$ are zero, so $E = N W P_{zw}^*$ does not depend on the sensor-noise spectrum V , while $M = P_{zw} W P_{zw}^* + V$.

Proposition 8. *Consider a stable plant with $H \succeq 0$ and parameterization $Q = D$. For a fixed Q , the cost (44) is affine in V , with*

$$\delta \bar{J} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{2} \text{Tr} \left(Q^* \tilde{H} Q \delta V \right) d\omega. \quad (59)$$

Consequently, the optimal causal cost, written $\bar{J}_c(V) = \min_Q \bar{J}(Q; V)$, is concave and non-decreasing in V , and wherever it is differentiable its sensitivity to sensor i 's noise level is

$$\frac{\partial \bar{J}_c}{\partial V_{ii}} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{2} \left[Q_c^* \tilde{H} Q_c \right]_{ii} d\omega \geq 0, \quad (60)$$

where Q_c is the causal optimum. The same holds for the pointwise optimum with Q_{pw} .

Proof: Appendix G, p. 42.

Corollary 9 (Value of a sensor upgrade). *Under the assumptions of Proposition 8, let Q_c be the causal optimum for the noise spectrum V and let V' be any other noise spectrum (not necessarily close to V , and possibly frequency dependent). Then*

$$\bar{J}_c(V) - \bar{J}_c(V') \geq \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{2} \text{Tr} \left(Q_c^* \tilde{H} Q_c (V - V') \right) d\omega. \quad (61)$$

The right-hand side is exactly the improvement obtained by keeping the current controller; re-optimizing for V' can only add to it.

Proof: Appendix G, p. 43. For an upgrade ($V' \preceq V$) the bound is non-negative and is computed from the current design alone.

The quantity (60) is the marginal value of sensor i 's accuracy: the energy, weighted by the plant's cost curvature, with which the optimal controller uses that sensor. It is computable from frequency responses and measured spectra, needs no state-space model, and is available at no extra cost once Q_c is known. Some uses:

- **Accuracy allocation.** With candidate sensors of precision $p_i = 1/V_{ii}$ (for white sensor noise) and a price $c_i(p_i)$ (e.g. linear in precision, or reflecting averaging time or beam count), minimize $\bar{J}_c(p) + \sum_i c_i(p_i)$. The gradient is $\partial \bar{J}_c / \partial p_i = -p_i^{-2} \partial \bar{J}_c / \partial V_{ii}$, so a projected-gradient or proximal method applies; a sensor whose optimal precision is zero is simply not installed, which gives placement as a by-product.
- **Placement with a fixed cost per sensor.** Removing a sensor is the limit $V_{ii} \rightarrow \infty$. Greedy addition or removal with exact marginal costs mirrors the set-function approaches of [40, 41]; the pointwise bound (Proposition 3 with the augmented W) gives a cheap lower bound for screening candidate sets, and its sensing-gap term (31) identifies disturbance directions that no candidate set can see.
- **Value of preview.** A sensor that sees the disturbance τ seconds early enters P_{zw} with a factor $e^{s\tau}$. The causality cost $\frac{1}{2} \| [F]_- \|^2$ as a function of τ (and of the preview sensor's noise, via (60)) prices a lidar, a wave-measurement buoy, or a forecast subscription before it is bought.

Proposition 8 gives concavity in the noise spectrum V . For the allocation problem it is the reciprocal variable that matters, and in the state-space case the corresponding convexity can be proved.

Proposition 10. *Let $\Sigma_*(p)$ be the stabilizing solution of the filter Riccati equation $A\Sigma + \Sigma A^T + BB^T - \Sigma C^T \text{diag}(p) C \Sigma = 0$, for precisions $p \geq 0$ such that it exists, and let $G \succeq 0$. Then $p \mapsto \text{Tr}(G \Sigma_*(p))$ is convex. In particular the latency-aware cost (99), which is $\text{Tr}(PBB^T) + \text{Tr}(G \Sigma_*(p))$ with $G = e^{A^T \tau} K^T R K e^{A\tau} \succeq 0$, is convex in the precisions, and so is the allocation problem $\min_{p \geq 0} \bar{J}_c(p) + \sum_i c_i(p_i)$ for convex prices c_i .*

(Proof: Appendix G, p. 43; the argument is the information-form relaxation used by [25] for the estimation problem.)

This explains the convexity observed numerically in Section 7.1, and it is what makes the price-per-precision sweeps there and in Section 7.2 reliable. It says nothing about placement with a fixed cost per sensor, which stays combinatorial.

Numerical check (`src/verify_sensors.py`): for the plant of Figure 2 with sensors $z = x + v$, $V = \text{diag}(0.04, 0.09)$, (60) evaluated at the LQG Youla parameter gives 1.499 and 1.019, against 1.500 and 1.022 from finite differences of the exact LQG cost. $\bar{J}_c$ is concave along V_{11} (all second differences negative), and removing sensor 2 (resp. 1) raises the cost from 2.866 to 3.225 (resp. 2.970): the less accurate velocity sensor is worth more here.

6 Software: the `freqctrl_v2` package

The companion Python package `freqctrl_v2` (in `src/`) implements the design end to end.

- **Input.** Either a state-space plant $\dot{x} = Ax + B_u u + B_w w$, $z = C_z x + D_{zw} w + v$ (stable or unstable, any number of inputs and outputs), or frequency responses of a stable plant on a grid. Disturbance and sensor-noise spectra $W(\omega)$, $V(\omega)$ are arbitrary functions of frequency (e.g. measured spectra). The objective is either $y_a = y_b = [Q^{1/2}x; R^{1/2}u]$ or a general pair y_a, y_b ; an indefinite objective is detected and rejected (Proposition 2).
- **Parameterization.** For state-space input, the Youla parameterization (43) is built around an observer-based controller $u = -K_f \hat{x} + Qe$ with innovation $e = z - C_z \hat{x}$ (the architecture of Boyd and Barratt, Fig. 15.1), which is valid for unstable plants; for frequency-response input, $Q = D$.
- **Tuning.** Greedy pole selection in the weighted metric (58) (any size) or in whitened coordinates (57) (one input and one measurement), with pole refinement and optional validation-grid early stopping.
- **Output.** Every intermediate controller, together with the non-causal pointwise optimum and (for one input and one measurement) the Wiener-Hopf optimum of Theorem 6, and a nine-panel plot: cost per step, controller magnitude and phase, closed-loop impulse response, cost density, selected poles, the disturbance-to-output response (with the uncontrolled plant) in frequency and time, and the loop transfer function $L = P_{zu}C$.

Table 2 summarizes four demonstrations (`src/examples/demo_autotune_v2.py`) with a narrowband disturbance $W(\omega) = |G_d(i\omega)|^2$, $G_d(s) = 2\omega_0^2/(s^2 + 2\zeta\omega_0 s + \omega_0^2)$, $\omega_0 = 1.5$, $\zeta = 0.2$, for which the nominal (white-noise) LQG controller is not optimal. The exact causal optimum is LQG on the plant augmented with the shaping filter G_d . Figures 6 and 7 show two of the cases.

Case	nominal	greedy (fns)	validated	exact causal	pointwise
A: SISO, state space	14.213	10.282 (12)	10.283	10.283	8.766
B: SISO, freq. response	13.217	10.282 (16)	10.283	10.283	8.766
C: 2×2 masses	0.299	0.206 (10)	0.206	0.206	0.131
D: unstable (pole +1)	16.018	10.523 (10)	10.530	10.531	3.488

Table 2: Demonstrations of `freqctrl.v2.autotune` (weighted greedy, tuning grid $dt = 0.005$ s, validation grid $dt = 0.00125$ s). “Nominal” is the parameterization’s $Q = 0$ controller (white-noise LQG for state-space input; open loop for frequency-response input). “Exact causal” is augmented LQG. For B, the plant is given only through its frequency response. The remaining differences between the validated and exact costs ($\leq 7 \times 10^{-4}$) are within the validation grid’s own discretization error.

7 Worked examples: latency, sensor placement and sensor accuracy

This section applies the method to three problems in which sensing is a design choice. Each example has a state-space model, so every frequency-domain result can be checked against an exact optimum (Appendix H for latency, LQG with augmented shaping filters otherwise). The frequency-domain computations use only frequency responses and spectra; the state-space models serve only as a check. The examples therefore validate the method and illustrate how Propositions 6 and Corollary 9 would be used. They do not show an advantage on a problem where no state-space model exists (for example measured frequency responses and ambient PMU spectra), which is where the method is intended to matter.

Where these results stand relative to prior work is set out in Section 9; in short, latency-aware LQG, sensor selection by closed-loop performance, and the pricing of measurement effort are all established, and what is being illustrated here is the frequency-domain route to them and the sensing results of Section 5.

Everything below is reproducible from the repository: the three demonstrations and the models they use are in `src/examples/`, and the checks are in `src/tests/`.

7.1 A stylized two-area system

Ten classical machines in two areas, small enough that every PMU subset can be enumerated. The inter-area mode is at 0.43 Hz with 3.9% damping, two inverter-based resources are the actuators, and a candidate PMU at each bus reports machine frequency with noise and latency τ . The objective charges tie-line swings, frequency deviations from the centre of inertia, and actuator effort. Appendix J.1 states the problem in full. Open loop the cost is 7.666; with full state information, no noise and no latency, 1.379.

Latency. For the PMU set $\{2, 8, 10\}$ (the best set of three, below), the frequency-domain tuner is given the frequency responses with the factor $e^{-s\tau}$ and the load spectra. Table 3 compares its validated cost with the exact optimum (99). The agreement is within 0.06% at every latency, with 24–26 basis functions. An LQG controller designed for zero latency and then operated with latency is unstable already at $\tau = 0.1$ s for this

PMU set (stability checked with Padé models of the delay; orders 6, 8 and 12 give the same rightmost closed-loop eigenvalue to five digits). Its cost rises from 1.983 at $\tau = 0$ to 3.564 at $\tau = 0.08$ s (closed-loop spectral abscissa -0.025 , i.e. close to instability), while the latency-aware optimum rises only to 2.175. The non-causal bound (1.330) does not depend on τ , because a pure delay is all-pass. The gap between the causal optimum and this bound grows with latency, so latency makes the cost of causality larger (Figure 8d). Figure 8c shows the response of a tie line to a load step. Figure 9 shows the tuner’s controller, closed-loop responses and loop gain next to the exact optimum.

τ (s)	tuned (validated)	basis functions	exact (99)	exact controller on grid	LQG ignoring latency
0.0	1.9845	24	1.9838	1.9833	1.9833
0.1	2.2290	26	2.2282	2.2275	unstable
0.2	2.5083	26	2.5080	2.5072	unstable
0.3	2.7955	24	2.7940	2.7933	unstable

Table 3: Two-area system, PMUs at buses 2, 8, 10. “Validated”: the tuned rational controller evaluated on a grid four times finer than the tuning grid. The exact controller’s cost on the grid differs from (99) by the $O(\Delta t)$ discretization error.

Placement with a fixed cost per PMU. Evaluating (99) for all 1023 subsets takes 4 s. Figure 10(a,b) and Table 4 compare the exhaustive optimum with forward greedy selection, with one-shot screening by the Proposition 8 marginal values at the all-PMU design, and with a common heuristic (largest magnitude of the inter-area mode shape). The optimal sets are not nested: the best single PMU is at bus 10, the best pair is $\{2, 10\}$ and the best set of four is $\{1, 3, 8, 10\}$. Greedy selection is optimal for $k \leq 3$ and $k \geq 8$, and at most 0.9% worse otherwise. The two screening rules are within 3.1% (Proposition 8) and 3.7% (mode shape) of the optimum. The differences between good choices are small; the main value of the evaluation is to avoid bad ones: the worst sets of three and four PMUs (all in one area) cost 52% and 55% more than the best.

k	exhaustive optimum (buses)	greedy	screening by Prop. 8	mode-shape heuristic	median	worst
1	3.394 (10)	3.394	3.435	3.435	3.478	3.815
2	2.376 (2, 10)	2.376	2.395	2.380	2.540	3.733
3	2.228 (2, 8, 10)	2.228	2.285	2.235	2.344	3.387
4	2.119 (1, 3, 8, 10)	2.137	2.184	2.198	2.244	3.275
5	2.066	2.078	2.123	2.098	2.147	3.221
6	2.023	2.026	2.026	2.056	2.080	2.296
10	1.936	1.936	1.936	1.936	1.936	1.936

Table 4: Two-area system, latency 0.1 s: exact optimal cost for k PMUs chosen by each rule; median and worst over all $\binom{10}{k}$ subsets.

Accuracy with a price per unit precision. Proposition 8 evaluated at the exact optimum on the frequency grid agrees with the Riccati adjoint (and with finite differences) to 1.9% on the tuning grid and 0.47% on the four times finer grid, as expected for an $O(\Delta t)$ error. The marginal values (Figure 10c) are largest at the far ends of the two

areas (buses 1, 9, 10) and smallest near the tie lines (buses 4, 5, 6). Resolved in frequency (Figure 10d, per decade, so that areas compare contributions), most of the value lies between about 1 and 3 Hz, the band of the local modes (1.15–2.7 Hz), with a small contribution below 0.1 Hz. The inter-area band (0.43 Hz) contributes little at this design point (all ten PMUs). A heuristic that places or upgrades PMUs for the inter-area mode alone would not see this. This is an example of the kind of information that a placement-only analysis does not provide. With a linear price c per unit precision, minimizing $\bar{J}_c(p) + c \sum_i p_i$ over $p \geq 0$ (L-BFGS-B with the adjoint gradient) gives the same optimum from six random starts (objective spread below 10^{-13}). Along 300 random segments in the precision space, all second differences of $\bar{J}_c$ were positive, as Proposition 10 requires. A linear price spreads precision over almost all PMUs (Figure 10e): at $c = 0.3$ nine of ten buses receive nonzero precision, and at lower prices all ten do. Diminishing returns in each PMU favour spreading. Choosing a few locations becomes the decision only when each PMU also carries a fixed installation cost, which is the combinatorial problem above.

7.2 IEEE 39-bus (New England) system, classical-machine model

The IEEE 39-bus (New England) system, with network, loads, power flow, machine and governor data from the case distributed with ANDES 2.0, reduced to a classical-machine model without exciters or power system stabilizers. In that model the governors damp the inter-area mode (0.68 Hz, the external-system equivalent at bus 39 against the rest) to 13.5%, so this example is about wide-area frequency and oscillation regulation with noisy, delayed PMUs rather than about rescuing a poorly damped mode. Two actuators act on machines at buses 33 and 38; a candidate PMU at each of the 39 buses reports bus frequency through a 0.05 s filter. Appendix J.2 states the problem, the data and the simplifications in full. Open loop the cost is 19.157; with full information, no noise and no latency, 4.613; with PMUs at all 39 buses and $\tau = 0.1$ s, 6.990. **Latency.** With PMUs at buses 20, 38, 39 (the best set of three, below), the frequency-domain tuner (Figure 12) reaches 8.7418 on the validation grid with 16 basis functions at $\tau = 0.1$ s. The exact optimum (99) is 8.7406, and the exact controller on the same grid gives 8.7376. At $\tau = 0.3$ s the tuner reaches 12.6074 against 12.6048. The LQG controller designed for zero latency costs 7.98 at $\tau = 0$, 10.76 at $\tau = 0.1$ s (23% above the latency-aware optimum) and 78.0 at $\tau = 0.15$ s, and it is unstable from $\tau = 0.175$ s on (Figure 11c). The non-causal bound for this PMU set is 6.498 at every latency.

Placement with a fixed cost per PMU (Table 5, Figure 11b). Exhaustive search covered all sets of up to four PMUs (92 170 sets). The best sets are not nested:

- one PMU: bus 16;
- two: {20, 38};
- three: {20, 38, 39};
- four: {20, 23, 29, 39}.

Placement matters more than in the two-area system: the best set of three costs 27% less than the median set and 39% less than the worst. The simple rules are less reliable here:

- **Greedy selection** starts with the best single PMU (bus 16) and is then 12% and 14% above the optimum for two and three PMUs, because bus 16 is not part of the best pairs or triples.
- **One-shot screening by the Proposition 8 marginal values** at the all-PMU design picks neighbouring buses with correlated information (29, 38, 20, 28, 34, ...). For small sets it is poor: for two PMUs it is worse than the median random pair. From about six PMUs on it is within 3% of greedy, and at 11 and 12 PMUs it is slightly better.
- **The mode-shape heuristic** is good for two PMUs (1.4% above the optimum) but 15–19% above it for three and four.

k	exhaustive optimum (buses)	greedy	screening by Prop. 8	mode-shape heuristic	random median
1	13.518 (16)	13.518	14.628	14.766	14.259
2	10.327 (20, 38)	11.568	14.434	10.475	12.802
3	8.741 (20, 38, 39)	9.951	10.105	10.035	11.813
4	8.324 (20, 23, 29, 39)	8.395	10.022	9.881	11.260
6	–	7.929	8.142	9.599	10.361
8	–	7.644	7.772	9.217	9.649
12	–	7.327	7.307	8.970	8.574
39	6.990	6.990	6.990	6.990	6.990

Table 5: IEEE 39-bus model, latency 0.1 s: exact optimal cost for k PMUs chosen by each rule; random median over 200 random sets.

Accuracy with a price per unit precision. At the selected set, Proposition 8 on the frequency grid agrees with the Riccati adjoint to 0.4%, 0.2% and 0.04% for the three PMUs. With a linear price c per unit precision over all 39 candidates (L-BFGS-B with the adjoint gradient; about 38 ms per gradient), the optimal allocation becomes sparse as the price rises:

- $c = 0.003$: 10 buses receive nonzero precision, $\bar{J}_c = 5.94$;
- $c = 0.055$: 6 buses, $\bar{J}_c = 7.04$;
- $c = 1$: 5 buses, $\bar{J}_c = 9.56$ (Figure 11e).

In this instance the ranking by allocated precision recovers the exhaustive optimum for the fixed-cost problem where greedy selection does not. The four buses with the largest precision at $c = 0.055$ and $c = 0.234$ are {20, 23, 29, 39}, the best set of four. The three largest at $c = 1$ are {20, 38, 39}, the best set of three. This is the idea behind convex relaxations of sensor selection [19], here with the exact closed-loop cost including latency.

We report it as an observation on one instance: we have not shown that the continuous problem is convex, beyond the numerical probe in Section 7.1, or that the recovery holds in general.

7.3 Sensor fusion with frequency-dependent sensor noise

A platform on a passive isolator, driven by ground motion, with two dissimilar sensors: a relative sensor with white noise, which sees the ground motion as if it were noise, and an inertial sensor whose noise rises steeply below its corner frequency. Neither is good everywhere, so the question is what the optimal controller does with them, and which sensor is worth improving. Appendix J.3 states the problem in full.

Results (Figure 13). Open loop: 0.838. Exact optimum: 0.4983 (its controller evaluated on the tuning grid: 0.4982). Weighted greedy tuning reaches 0.4983 on the validation grid with 24 basis functions (early stopping). With one sensor only, the exact optimum is 0.680 (relative) or 0.543 (inertial), so blending is worth about 8% over the better single sensor. The tuned controller reproduces the exact blending (panel b): the relative sensor is used with a high gain at low frequency, where it holds the platform to the ground and the inertial sensor is noisy, and the inertial sensor’s gain peaks between about 0.1 and 5 Hz. Nothing about complementary filters was imposed; the blend follows from the noise spectra.

Which upgrade to buy. Panel (c) shows the Proposition 8 density $g_i(\omega) = V_{ii}(\omega) \partial \bar{J}_c / \partial V_{ii}(\omega)$. It is the share of the cost that would disappear per unit relative noise reduction, frequency by frequency. Panel (d) compares, for four candidate upgrades, the lower bound (61), computed from the current design alone, with the exact decrease after redesign:

upgrade	bound (61)	current controller, re-evaluated	exact, after redesign
relative-sensor noise $\div 10$	0.0209	0.0209	0.0807
inertial noise floor $\div 10$	0.0466	0.0466	0.1468
inertial corner $0.5 \rightarrow 0.17$ Hz	0.0507	0.0507	0.2606
both inertial upgrades	0.0517	0.0517	0.3462

The second column checks that the cost of a fixed controller is exactly affine in V (Proposition 8). The bound holds in every case and ranks the three single upgrades in the same order as the exact values. It is conservative, by a factor of 3 to 7 here: the gap is the value of redesigning the controller for the better sensor, which a first-order analysis of the current design cannot see. So the bound is useful for ruling options in (a guaranteed minimum return) and for a first ranking, not for estimating the full return.

7.4 Fixed-structure PID on a benchmark plant

The Wood–Berry distillation column, a 2×2 plant of first-order lags with delays, is the example used by Boyd, Hast and Åström for MIMO PID tuning by iterated LMI restriction [4]. Their design minimizes a low-frequency sensitivity subject to peak constraints

on S , T and Q ; ours minimizes a spectral cost over the same PID structure, using the stationarity condition (38) with the admissible set $\{1, 1/s, s/(\tau s + 1)\}$. Appendix J.4 states both problems in full. The delays are exact on the grid and no state-space model is used; their published gains are reproduced as a check, and the gradient of Section 3 agrees with finite differences to 6×10^{-10} .

design	spectral cost	$\ S\ _\infty$	$\ T\ _\infty$	$\ Q\ _\infty$	$\ (P(0)\mathcal{I})^{-1}\ $
LMI restriction [4]	0.0177	1.406	1.398	0.739	2.25
spectral cost alone	0.0025	1.748	1.008	1.971	0.44
<i>but: two unstable closed-loop poles</i>					
spectral cost, peaks penalized, stability enforced	0.0148	1.400	1.135	0.738	2650
integrating disturbance, peaks penalized, stability enforced	—	1.400	1.201	0.738	4.07

Table 6: Wood–Berry column, MIMO PID. The two formulations optimize different things, so neither column of metrics ranks the designs; what the table shows is that the structure condition of Section 3 handles this plant (delays and all) directly from frequency responses, and what it costs to ignore the constraints that the LMI formulation carries.

What this shows. Table 6 summarizes the designs and Figure 14 in Appendix I compares them in frequency and in time. The framework computes the design: PID is one of its lag sets, the delays need no approximation, and the gradient is exact. But minimizing the spectral cost alone leaves the stabilizing set, as described in Section 3: along the segment from the published design to this minimizer, $\det(P_{zu}(0)\mathcal{I})$ passes through zero at about 2% of the way, two closed-loop poles cross into the right half-plane, and the cost rises only from 0.018 to 0.033 before falling to 0.003. With the peaks penalized and stability enforced by an encirclement test, the design is stable, meets all three constraints, and has a 16% lower spectral cost than the LMI design — but its low-frequency sensitivity is three orders of magnitude worse, because the spectral cost does not ask for tracking and the gradient pushes towards precisely the boundary where the integral action becomes singular. **Matching the LMI design by choosing the disturbance.** The mismatch in the last column is a statement about the objective, not about the method. Their objective is the coefficient in $S(s) \approx s(P_{zu}(0)\mathcal{I})^{-1}$ near the origin, so a disturbance that is concentrated at low frequency penalizes exactly that quantity: with $W(\omega) = (\omega^2 + a^2)^{-1}$, an integrating disturbance, the integrand tends to $\|(P_{zu}(0)\mathcal{I})^{-1}\|_F^2$ as $\omega \rightarrow 0$. Optimizing this cost, with the peaks penalized and stability enforced as before, gives a design that sits essentially on top of theirs: gains $K_P = [0.1832, -0.0454; -0.0781, -0.0692]$ against their $[0.1750, -0.0470; -0.0751, -0.0709]$, all three constraints met ($\|S\|_\infty = 1.400$, $\|T\|_\infty = 1.201$, $\|Q\|_\infty = 0.738$, and zero encirclements), a marginally lower cost in the integrating-disturbance measure (151.60 against 151.97), and their own objective at 4.07 against 2.25 — the remainder being the control effort and mid-band response that our cost also charges and theirs does not. One practical point: this comparison has to be integrated on a logarithmically spaced grid. Their objective is an asymptotic statement as $s \rightarrow 0$, and the uniform grid used elsewhere in this paper has its lowest non-zero frequency at $2\pi/(N dt)$, far above the band where that asymptote governs; on the uniform grid the same weighting appears to suc-

ceed while leaving $\|(P_{zu}(0)\mathcal{I})^{-1}\|$ three orders of magnitude too large. Nothing in the method requires the uniform grid except the causal projection of Section 4.

For a PID design with peak specifications the LMI restriction remains the more direct tool, since it carries the constraints natively; what this framework adds is the cost and its gradient computed from data with exact delays, and the freedom to state the design goal as a disturbance spectrum rather than as a norm.

7.5 Scope, scaling, and relation to integer-programming placement

What the cost depends on, and what it does not. Once frequency responses are available, nothing in the design sees the plant’s state dimension: every operation at a frequency acts on matrices whose size is set by the numbers of inputs, sensors and objective outputs. The 39-bus model has 98 states against the two-area model’s 39, yet each greedy tuning run took about 124s against 210–300s — the cost of a run is set by the number of basis functions (16 against 24–26) and by $n_u n_z$, not by n_x — and the Wood–Berry column, whose delays make it infinite-dimensional, needed no approximation at all. We state this as a property rather than a contribution: every method that works from frequency responses shares it, and it moves the difficulty rather than removing it. The order of the *controller* still grows with what the plant does, since lightly damped modes need basis poles near them; and the frequency grid must resolve the plant’s sharpest features and, as Section 7.4 showed, whatever band the objective is about.

The classical PMU placement problem chooses the fewest PMUs that make every bus topologically observable (a PMU observes its bus and its neighbours), which is a set-covering integer program. It scales to thousands of buses and handles hard constraints (budgets, redundancy, zero-injection buses), but every covering set counts as equally good. The analysis here answers a different question: how good a given set is for a given closed-loop objective, including noise, latency and measured spectra, and what accuracy is worth. In the two-area system, the best and worst sets of three PMUs differ by 52% in cost. The two approaches are complementary. Covering and redundancy constraints can be imposed on the candidate sets, and the cost used here can serve as the objective, or its gradient can drive a relaxation.

Computational cost, measured on a laptop (single process):

- **Exact evaluation of one PMU set** (state-space check, (99)): one filter Riccati equation. 4 ms for the 39-state two-area model; 22 ms for the 98-state 39-bus model. Exhaustive search is therefore limited to small sets or few candidates: in the 39-bus model, all 9 139 sets of three PMUs took 232 s, and all sets of four (82 251) took 3026 s.
- **Greedy selection:** $k \times$ (number of candidates) evaluations. It took 37 s for all sizes $k = 1, \dots, 39$ in the 39-bus model.
- **Accuracy gradient for all candidates:** one Riccati and one Lyapunov equation (the adjoint of Proposition 8), 38 ms for 39 candidates. Its cost hardly depends on the number of candidates, so accuracy allocation scales best.

- **Frequency-domain tuning** (Section 4): memory and time grow with the number of frequency points and as $(n_u n_z)^2$ per point, so the present implementation suits controllers with few inputs and outputs ($n_u n_z \lesssim 30$). A 2×3 controller took 3–5 minutes per run in the two-area example when run without competing jobs. Computing frequency responses of large models is not the bottleneck: they are obtained from one eigendecomposition in $O(Nn)$ operations.

For a realistic wide-area damping design (a few actuators, tens of candidate PMUs, reduced models with hundreds of states) these costs are practical. A full interconnection would need model reduction, sparse solvers, or a relaxation layer; we have not tested that.

8 Where this should be useful

The framework earns its keep where disturbances are spectral and often measured rather than modelled, where the plant is known through frequency responses, where the objective is a product of two different signals, or where sensing and preview are themselves design choices. The following are candidates, each with the result it would use. None has yet been carried out on a realistic model; the examples of Section 7 are the closest we have come.

1. **Wave energy converters and arrays.** Force $\times$ velocity, bounded below by passivity plus losses [26]. The causal optimum comes from hydrodynamic frequency responses without rational fits of the radiation kernels, and Proposition 8 prices up-wave elevation sensors and wave forecasts against the cost of causality [33, 13].
2. **Wind turbines, fixed and floating.** Power against tower and blade loads under a turbulence spectrum, with lidar preview [15] or wave feedforward [11]. The lidar’s range, beam count and averaging are sensor parameters that (60) prices directly; pitch and torque loops are fixed-structure designs in the sense of (36).
3. **Power grids and inverter fleets.** Inter-area damping and coherency [7, 31], or objectives such as $\Delta P \cdot \Delta f$. Section 7 does this one.
4. **Precision isolation and sensor fusion.** Blending relative and inertial sensors from measured noise spectra [39], which Section 7.3 treats; the same structure appears in lithography stages and disk-drive servos.
5. **Adaptive optics.** The servo-lag error is a cost of causality, and temporal prediction reduces it by a factor of up to 1.77 in variance for von Kármán turbulence [6]; wavefront-sensor configuration trades exposure against noise, which is Proposition 8.
6. **Active noise and vibration control.** Processing latency limits attenuation, and a reference microphone further upstream is preview; both are quantities this framework computes rather than assumes.

7. **Energy systems with price-like objectives.** An energy bill is a cross-spectrum between price and power, and degradation supplies the losses that make the problem bounded. Forecast value follows from the cost of causality. Constraints such as state of charge are outside the linear theory and would need a layer on top.
8. **Vibration energy harvesting, structures and flows.** Harvested power is a sign-indefinite objective [18]; semi-active damping of buildings dissipates a force $\times$ velocity product; sensor and actuator placement in flow control can start from resolvent or measured data [29, 27].

9 Related work

Most of the ingredients used here are established, and the purpose of this section is to say which, and where the treatment above differs. The repository’s review file records the comparison in more detail, including which sources have been read in full.

Frequency-domain optimal control. For $y_a = y_b$ the pointwise solution is the non-causal Wiener regulator written in internal-model-control coordinates [29], and the recovery $C = (I + DP_{zu})^{-1}D$ is the Youla parameterization [43, 21, 44, 5, 37]. Theorem 6 is the classical Wiener–Hopf solution in those coordinates; Martini et al. solve the positive-definite case numerically from resolvent data and use the non-causal kernel as a performance bound [29]. The gap between a causal design and a clairvoyant one is also the subject of regret-optimal and competitive-ratio control [34, 14].

Power objectives. The pointwise optimum for a force-times-velocity objective is classical for arrays of interacting oscillators: Falnes gives the optimum-velocity condition, the quadratic expansion about it, the closed-form maximum with and without friction losses, and the observation that when the radiation-resistance matrix is singular the optimal velocity is ambiguous while the optimal power is unique [8, 9]. Scruggs et al. carry the same objective into a causal design for a wave energy converter, noting that the performance functional is sign-indefinite, obtaining the pointwise optimum frequency by frequency, solving the causal problem as a Riccati equation after fitting finite-dimensional models of both the plant and the sea spectrum, and exhibiting an estimation-error penalty in the output-feedback cost of the same kind as the sensing gap in Proposition 3 [36]; existence rests there on positive realness of the driving-point impedance [35]. Relative to these, Sections 2 and 4 take arbitrary output pairs and arbitrary sensing and actuation structure, give the optimal cost in closed form with its two parts, replace the passivity assumption by a pointwise test on the data, and avoid rational fits. The wider wave-energy literature on complex-conjugate control and its causal approximations is surveyed in [33, 30, 26, 13].

Structure constraints and finite bases. Tuning fixed-structure controllers from frequency-response data is established practice [20, 4], as is the study of the resulting non-convex

landscapes [10, 45]. The constant-gain gradient of Proposition 5 is the one attributed to Levine and Athans [23]; what Section 3 adds is that it, and the causal, preview and PID conditions, are the same condition on different lag sets. Expanding the Youla parameter in a finite basis around a nominal observer-based controller is Boyd and Barratt’s Ritz approximation [5, §15], and the behaviour of orthonormal rational bases under pole choice is well studied [17, 24]. The greedy rule of Section 4 is the maximal selection principle of adaptive Fourier decomposition [32, 12, 1], here applied in the weighted metric of the control problem so that it works for several inputs and outputs without matrix spectral factorization.

Sensing. Choosing sensors by estimation or closed-loop performance has a large literature, including convex relaxation [19], supermodularity and greedy guarantees [40, 41], sensing regularized by information [38], and sparse placement from data [27]. Pricing measurement effort inside the problem goes back to [28, 3]. In continuous time, sensors are scheduled across plants with a cost per unit time, and the relaxed problem is convex in the information matrix [25]; they may also be switched, with switching costs [22]; in discrete time a measurement budget is spread over a horizon, with quality traded against quantity [2]. These are estimation problems, with the estimation error and the price of measuring in the cost and no closed-loop term. Proposition 10 is the information-form argument of [25] applied to the closed-loop cost, and is not claimed as new. What we have not found stated is Proposition 8 in a form computable from frequency responses and measured spectra, its resolution in frequency, and the upgrade bound of Corollary 9.

What we could not check. Several works bearing on the attributions above were not obtained: [28] and [3], the origin of measurement pricing; [23], whose condition we attribute but have not read; [35]; and [16, 43]. Three works that cite [28] describe it as solving sensor selection in discrete time by complete enumeration over open-loop schedules [22, 25, 2], which would make it a combinatorial result rather than a sensitivity, but the original should be read before relying on that.

10 Outlook

Several things are missing or unfinished.

- Theorem 6 for several inputs needs matrix spectral factorization of $\tilde{H}$ on a grid (e.g. Wilson’s method), not only the scalar cepstral one used here; and the causal problem for power-type objectives needs $\tilde{H} \succ 0$, i.e. passivity plus losses.
- Preview: replace $[\cdot]_+$ by the projection onto lags $t \geq -\tau$, and plot the cost against the preview time τ .
- Sensing selection with the controller coupled: Section 7 gives first examples (placement, accuracy, latency); open are relaxations for large candidate sets and tests on measured data.

- Structure selection, as opposed to structure-constrained optimization: the lag-set condition scores a given structure, but choosing among structures is open.
- Decentralization. Everything here assumes one controller with one measurement vector, so that the closed loop is affine in the Youla parameter. With several controllers each seeing part of the system that can fail, and the test is whether one controller’s influence on another’s measurements is matched by information reaching that controller — which is what quadratic invariance formalizes. Structures that are quadratically invariant with respect to the plant [42] keep that property and should admit the same treatment, with the admissible set describing the decentralization as well as the lag structure; structures that are not quadratically invariant do not, and are outside this framework.

References

- [1] D. Alpay, F. Colombo, T. Qian, I. Sabadini, Adaptive orthonormal systems for matrix-valued functions, arXiv:1511.07834, 2015.
- [2] A. Aspeel, A. Legay, R. M. Jungers, B. Macq, Optimal measurement budget allocation for Kalman prediction over a finite time horizon by genetic algorithms, EURASIP Journal on Advances in Signal Processing 2021:39 (read; copy in `refs/`).
- [3] M. Athans, On the determination of optimal costly measurement strategies for linear stochastic systems, Automatica 8(4):397–412, 1972 (not obtained; cited as reference [6] of [25]).
- [4] S. Boyd, M. Hast, K. J. Åström, MIMO PID tuning via iterated LMI restriction, Int. J. Robust Nonlinear Control 26:1718–1731, 2016.
- [5] S. Boyd, C. Barratt, Linear Controller Design: Limits of Performance, Prentice Hall, 1991.
- [6] N. Doelman, The minimum of the time-delay wavefront error in adaptive optics, arXiv:1906.03128, 2020.
- [7] F. Dörfler, M. R. Jovanović, M. Chertkov, F. Bullo, Sparsity-promoting optimal wide-area control of power networks, IEEE Trans. Power Systems, 2014 (arXiv:1307.4342).
- [8] J. Falnes, Radiation impedance matrix and optimum power absorption for interacting oscillators in surface waves, Applied Ocean Research 2:75–80, 1980.
- [9] J. Falnes, Ocean Waves and Oscillating Systems: Linear Interactions Including Wave-Energy Extraction, Cambridge University Press, 2002 (§6.4 read; copy in `refs/`; it reproduces the results of [8]).

- [10] I. Fatkhullin, B. Polyak, Optimizing static linear feedback: gradient method, arXiv:2004.09875, 2020.
- [11] A. Fontanella, M. Al, J.-W. van Wingerden, M. Belloli, Model-based design of a wave-feedforward control strategy in floating wind turbines, *Wind Energy Science* 6:885–901, 2021.
- [12] Y. Gao, M. Ku, T. Qian, Fast algorithm of adaptive Fourier series, arXiv:1711.08604, 2017.
- [13] D. García-Violini, Y. Peña-Sanchez, N. Faedo, J. V. Ringwood, An energy-maximising linear time invariant controller (LiTe-Con) for wave energy devices, *IEEE Trans. Sustainable Energy* 11(4):2713–2721, 2020.
- [14] G. Goel, B. Hassibi, Regret-optimal estimation and control, arXiv:2106.12097, 2021.
- [15] F. Guo, D. Schlipf, P. W. Cheng, Evaluation of lidar-assisted wind turbine control under various turbulence characteristics, *Wind Energy Science* 8:149–171, 2023.
- [16] B. Hassibi, A. H. Sayed, T. Kailath, *Indefinite-Quadratic Estimation and Control*, SIAM, 1999.
- [17] P. S. C. Heuberger, P. M. J. Van den Hof, O. H. Bosgra, A generalized orthonormal basis for linear dynamical systems, *IEEE Trans. Automatic Control* 40(3):451–465, 1995.
- [18] A. Haji Hosseinloo, T. L. Vu, K. Turitsyn, Optimal control strategies for efficient energy harvesting from ambient vibrations, arXiv:1508.04163, 2015.
- [19] S. Joshi, S. Boyd, Sensor selection via convex optimization, *IEEE Trans. Signal Processing* 57(2):451–462, 2009.
- [20] A. Karimi, C. Kammer, A data-driven approach to robust control of multivariable systems by convex optimization, *Automatica* 85:227–233, 2017.
- [21] V. Kučera, *Discrete Linear Control: The Polynomial Equation Approach*, Wiley, 1979.
- [22] H. W. J. Lee, K. L. Teo, A. E. B. Lim, Sensor scheduling in continuous time, *Automatica* 37:2017–2023, 2001 (read; copy in `refs/`).
- [23] W. S. Levine, M. Athans, On the determination of the optimal constant output feedback gains for linear multivariable systems, *IEEE Trans. Automatic Control* 15(1):44–48, 1970 (not obtained; the form of their gradient and necessary condition used here is the one standard in the static output-feedback literature, and is checked numerically in `src/verify_structured.py`).

- [24] J. Li, Y. Lu, Y. Mo, J. Chen, Identification with orthogonal basis functions: convergence speed, asymptotic bias, and rate-optimal pole selection, arXiv:2512.21096, 2026.
- [25] J. Le Ny, E. Feron, M. A. Dahleh, Scheduling Kalman filters in continuous time, American Control Conference, 2009 (read; copy in `refs/`).
- [26] C. H. Ligeikis, J. T. Scruggs, Causal, stochastic MPC for wave energy converters, arXiv:2305.04094, 2023.
- [27] K. Manohar, B. W. Brunton, J. N. Kutz, S. L. Brunton, Data-driven sparse sensor placement for reconstruction, IEEE Control Systems Magazine, 2018 (arXiv:1701.07569).
- [28] L. Meier, J. Peschon, R. M. Dressler, Optimal control of measurement subsystems, IEEE Trans. Automatic Control AC-12(5):528–536, 1967 (not obtained; cited by [25, 2, 22] as the origin of measurement-cost formulations, and described in [22] as solving discrete-time sensor selection by complete enumeration over open-loop schedules; pages and issue as given in the reference list of [22]).
- [29] E. Martini, J. Jung, A. V. G. Cavalieri, P. Jordan, A. Towne, Resolvent-based tools for optimal estimation and control via the Wiener-Hopf formalism, arXiv:2201.00361, 2022.
- [30] R. Nie, J. Scruggs, A. Chertok, D. Clabby, M. Previsic, A. Karthikeyan, Optimal causal control of wave energy converters in stochastic waves: accommodating nonlinear dynamic and loss models, Int. J. Marine Energy 15:41–55, 2016.
- [31] B. K. Poolla, S. Bolognani, F. Dörfler, Optimal placement of virtual inertia in power grids, IEEE Trans. Automatic Control, 2017 (arXiv:1510.01497).
- [32] T. Qian, Y. B. Wang, Adaptive Fourier series: a variation of greedy algorithm, Advances in Computational Mathematics 34:279–293, 2011.
- [33] J. V. Ringwood, Wave energy control: status and perspectives 2020, IFAC World Congress, 2020.
- [34] O. Sabag, S. Lale, G. Goel, B. Hassibi, Regret-optimal LQR control, arXiv:2105.01244, 2023.
- [35] J. T. Scruggs, On the causal power generation limit for a vibratory energy harvester in broadband stochastic response, Journal of Intelligent Material Systems and Structures 21:1249–1262, 2010 (not obtained; cited as reference [33] of [36] for the positive-realness condition, the positivity of the optimal power, and the vanishing-noise limit).
- [36] J. T. Scruggs, S. M. Lattanzio, A. A. Taflanidis, I. L. Cassidy, Optimal causal control of a wave energy converter in a random sea, Applied Ocean Research 42:1–15, 2013 (doi:10.1016/j.apor.2013.03.004; read in full, copy in `refs/`).

- [37] S. Skogestad, I. Postlethwaite, *Multivariable Feedback Control: Analysis and Design*, Wiley, 1996.
- [38] T. Tanaka, H. Sandberg, SDP-based joint sensor and controller design for information-regularized optimal LQG control, IEEE CDC, 2015 (arXiv:1503.01848).
- [39] T. T. L. Tsang, T. G. F. Li, T. Dehaeze, C. Collette, Optimal sensor fusion method for active vibration isolation systems in ground-based gravitational-wave detectors, *Class. Quantum Grav.*, 2022 (arXiv:2111.14355).
- [40] V. Tzoumas, A. Jadbabaie, G. J. Pappas, Sensor placement for optimal Kalman filtering: fundamental limits, submodularity, and algorithms, American Control Conference, 2016 (arXiv:1509.08146).
- [41] V. Tzoumas, L. Carlone, G. J. Pappas, A. Jadbabaie, LQG control and sensing co-design, IEEE Trans. Automatic Control, 2020 (arXiv:1802.08376).
- [42] M. Rotkowitz, S. Lall, A characterization of convex problems in decentralized control, IEEE Trans. Automatic Control 51(2):274–286, 2006 (details to be checked against the original).
- [43] D. C. Youla, J. J. Bongiorno, H. A. Jabr, Modern Wiener-Hopf design of optimal controllers, Part I: the single-input case, IEEE Trans. Automatic Control 21:3–14, 1976; D. C. Youla, H. A. Jabr, J. J. Bongiorno, Part II: the multivariable case, *ibid.* 21:319–338, 1976 (author order and pages as listed by Kučera, ICINCO 2024; to be checked against the originals).
- [44] G. Zames, Feedback and optimal sensitivity: model reference transformations, multiplicative seminorms, and approximate inverses, IEEE Trans. Automatic Control 26(2):301–320, 1981.
- [45] Y. Zheng, Y. Tang, N. Li, Analysis of the optimization landscape of linear quadratic Gaussian (LQG) control, arXiv:2102.04393, 2021.

A Quadratic form, first-order term, and the gradient identity

Write $A \doteq P_{yau}$, $a \doteq P_{yaw}$, $B \doteq P_{ybu}$, $b \doteq P_{yb w}$, so that $H = A^*B + B^*A$, $N = A^*b + B^*a$, $J_0 = \text{Re Tr}(aWb^*)$ and (14) reads $J = \text{Re Tr}((AK + a)W(BK + b)^*)$. Two elementary facts are used throughout.

$$\overline{\text{Tr}(Q)} = \text{Tr}(Q^*), \quad \text{so } \text{Re Tr}(Q) = \text{Re Tr}(Q^*) \text{ for any square } Q, \quad (62)$$

$$\text{Re Tr}(QS) = \frac{1}{2}\text{Tr}((Q + Q^*)S) \text{ for any square } Q \text{ and Hermitian } S. \quad (63)$$

Fact (63) follows from (62): $\overline{\text{Tr}(QS)} = \text{Tr}(SQ^*) = \text{Tr}(Q^*S)$, so $\text{Re Tr}(QS) = \frac{1}{2}(\text{Tr}(QS) + \text{Tr}(Q^*S))$.

Proof of (18). Expanding,

$$(AK + a)W(BK + b)^* = AKWK^*B^* + AKWb^* + aWK^*B^* + aWb^*. \quad (64)$$

For the first term, $\text{Re Tr}(AKWK^*B^*) = \text{Re Tr}(B^*AS)$ with the Hermitian $S \doteq KWK^*$, which equals $\frac{1}{2}\text{Tr}((B^*A + A^*B)S) = \frac{1}{2}\text{Tr}(HKWK^*)$ by (63). For the second term, $\text{Re Tr}(AKWb^*) = \text{Re Tr}(b^*AKW)$. For the third term, by (62), $\text{Re Tr}(aWK^*B^*) = \text{Re Tr}(BKWa^*) = \text{Re Tr}(a^*BKW)$. The second and third terms add to $\text{Re Tr}((b^*A + a^*B)KW) = \text{Re Tr}(N^*KW)$. The fourth term gives J_0 . This is (18). $\square$

Proof of (19). Substitute $K + \delta K$ into (18). The quadratic part gives

$$\frac{1}{2}\text{Tr}(HKWK^*) + \frac{1}{2}[\text{Tr}(H\delta KWK^*) + \text{Tr}(HKW\delta K^*)] + \frac{1}{2}\text{Tr}(H\delta KW\delta K^*). \quad (65)$$

The two terms in brackets are complex conjugates of each other, since $\overline{\text{Tr}(HKW\delta K^*)} = \text{Tr}(\delta KWK^*H) = \text{Tr}(H\delta KWK^*)$, using $H = H^*$ and $W = W^*$. So the bracket equals $2\text{Re Tr}(HKW\delta K^*)$. The linear part gives the additional term $\text{Re Tr}(N^*\delta KW) = \text{Re Tr}(W\delta K^*N) = \text{Re Tr}(NW\delta K^*)$ by (62). Adding up, the change is $\text{Re Tr}((HK + N)W\delta K^*) + \frac{1}{2}\text{Tr}(H\delta KW\delta K^*)$, which is (19). $\square$

Gradient identity. From (11), $C^*(I_{nu} - P_{zu}^*C^*)^{-1} = [(I_{nu} - CP_{zu})^{-1}C]^* = D^*$. Substituting into (8) and using (15), (16),

$$\mathcal{P}(C) = P_{zw}W(P_{zw}^*D^*H + N^*) = [(HDP_{zw} + N)WP_{zw}^*]^*. \quad (66)$$

Hence $\mathcal{P}(C) = 0$ if and only if (20) holds. $\square$

Youla relations. The two forms in (11) agree by the push-through identity $C(I - P_{zu}C) = (I - CP_{zu})C$. From $D = (I - CP_{zu})^{-1}C$, $D = C + CP_{zu}D = C(I + P_{zu}D)$, so $C = D(I + P_{zu}D)^{-1} = (I + DP_{zu})^{-1}D$, which is (13). $\square$

B Stationary controller and second-order condition

D_{opt} satisfies (20). Let $Z \doteq P_{zw}W^{1/2}$, so that $M = ZZ^*$ and $NWP_{zw}^* = NW^{1/2}Z^*$. Two standard Moore-Penrose identities are used:

$$Z^*(ZZ^*)^\dagger = Z^\dagger, \quad Z^\dagger ZZ^* = Z^*. \quad (67)$$

With $K_{\text{opt}} = D_{\text{opt}}P_{zw}$ and (23),

$$(HK_{\text{opt}} + N)WP_{zw}^* = -HH^\dagger NWP_{zw}^* M^\dagger M + NWP_{zw}^*. \quad (68)$$

By (67), $NWP_{zw}^* M^\dagger M = NW^{1/2}Z^*(ZZ^*)^\dagger ZZ^* = NW^{1/2}Z^\dagger ZZ^* = NW^{1/2}Z^* = NWP_{zw}^*$. So the right-hand side is $-HH^\dagger NWP_{zw}^* + NWP_{zw}^*$, which is zero by (22). $\square$

When $y_a = y_b$, $H = 2A^*A$ and $N = 2A^*a$, and the range of A^* equals the range of A^*A , so (22) holds automatically.

Proof of Lemma 1. Write $X = X_{y_a, y_b}$, $Y \doteq XWP_{zw}^*$ and $C^d = C_{\text{opt}}^d = -YQ^\dagger$. Note that $P_{zu}Y = P_{zu}XWP_{zw}^*$, so

$$M = Q + P_{zu}Y. \quad (69)$$

Step 1: a sufficient condition. For any D , $KWP_{zw}^* = DP_{zw}WP_{zw}^* = DM$, so $(HK + N)WP_{zw}^* = HDM + NWP_{zw}^*$. If

$$DM = -Y, \quad (70)$$

then $(HK + N)WP_{zw}^* = -HH^\dagger NWP_{zw}^* + NWP_{zw}^* = 0$ by (22), i.e. (20) holds.

Step 2: the Youla parameter of C^d . By (11) and the push-through identity, $D^d \doteq (I - C^d P_{zu})^{-1} C^d = C^d (I - P_{zu} C^d)^{-1}$, which exists because $I - P_{zu} C^d$ is invertible (and $\det(I - C^d P_{zu}) = \det(I - P_{zu} C^d)$). Let $R \doteq Q^\dagger P_{zu} Y$. Since $I - P_{zu} C^d = I + P_{zu} Y Q^\dagger$, and $Q^\dagger (I + P_{zu} Y Q^\dagger)^{-1} = (I + R)^{-1} Q^\dagger$ (push-through; $I + R$ is invertible because $\det(I + R) = \det(I + P_{zu} Y Q^\dagger)$), we get, using (69),

$$\begin{aligned} D^d M &= -Y Q^\dagger (I + P_{zu} Y Q^\dagger)^{-1} M = -Y (I + R)^{-1} Q^\dagger M = -Y (I + R)^{-1} (Q^\dagger Q + R) \\ &= -Y (I + R)^{-1} [(I + R) - (I - Q^\dagger Q)] = -Y + Y (I + R)^{-1} (I - Q^\dagger Q). \end{aligned} \quad (71)$$

Using $(I + R)^{-1} = I - (I + R)^{-1} R$ and $R = Q^\dagger P_{zu} Y$, the last term is

$$Y (I + R)^{-1} (I - Q^\dagger Q) = Y (I - Q^\dagger Q) - Y (I + R)^{-1} Q^\dagger P_{zu} [Y (I - Q^\dagger Q)]. \quad (72)$$

So (70) holds for D^d as soon as $Y (I - Q^\dagger Q) = 0$.

Step 3: $Y (I - Q^\dagger Q) = 0$. $I - Q^\dagger Q$ is the orthogonal projection onto the null space of Q , so it suffices that every null vector of Q is a null vector of Y . Write $Z = P_{zw} W^{1/2}$ as before. Then $Y = XW^{1/2} Z^*$ and $Q = (P_{zw} - P_{zu} X) W^{1/2} Z^*$, so the null space of Z^* is contained both in the null space of Y and in the null space of Q . By rank-nullity and (27), $\dim \text{null}(Q) = n_z - \text{rank } Q = n_z - \text{rank } Z^* = \dim \text{null}(Z^*)$, so the two null spaces are equal, and $\text{null}(Q) = \text{null}(Z^*) \subseteq \text{null}(Y)$. Hence $D^d M = -Y$, and by Step 1, C^d satisfies (20).

Step 4: same cost. Both D^d and D_{opt} satisfy (20), so they have the same cost by Appendix C.

Step 5: invertible M . Then $\text{rank } Z = n_z$, so (27) means Q is invertible and $Q^\dagger = Q^{-1}$. The loop is then automatically well posed, since $I - P_{zu} C^d = I + P_{zu} Y Q^{-1} = (Q + P_{zu} Y) Q^{-1} = M Q^{-1}$ by (69). From (70), $D^d = -Y M^{-1} = D_{\text{opt}}$, hence $C^d = C_{\text{opt}}$. For (28), $I + D_{\text{opt}} P_{zu} = I - Y M^{-1} P_{zu}$, and by Sylvester's determinant identity $\det(I - Y M^{-1} P_{zu}) = \det(I - P_{zu} Y M^{-1}) = \det((M - P_{zu} Y) M^{-1}) = \det Q / \det M$. $\square$

Proof of Proposition 2(a). Let D satisfy (20), $K = DP_{zw}$, and let D' be any other Youla parameter with $K' = D' P_{zw}$. Put $\delta K = K' - K = (D' - D) P_{zw}$ in (19):

$$J(K') = J(K) + \text{Re Tr} ((HK + N)WP_{zw}^* (D' - D)^*) + \frac{1}{2} \text{Tr} (H \delta K W \delta K^*). \quad (73)$$

The middle term is zero by (20). The matrix $S = \delta K W \delta K^*$ is positive semidefinite, and for $H \succeq 0$, $\text{Tr}(HS) = \text{Tr}(H^{1/2} S H^{1/2}) \geq 0$. Hence $J(K') \geq J(K)$. $\square$

Proof of Proposition 2(b). Let v be a unit vector with $v^*Hv < 0$. Since $M \neq 0$, there is a vector r with $r^*Mr > 0$. Starting from $D = 0$ (i.e. $C = 0, K = 0$), take $D_t = t vr^*$ for real t , so $K_t = t vr^* P_{zw}$. By (19) with $K = 0$ and $G(0) = NW$,

$$J(K_t) = J_0 + t \operatorname{Re} \operatorname{Tr} (NW P_{zw}^* r v^*) + \frac{t^2}{2} (v^* H v)(r^* M r), \quad (74)$$

using $\operatorname{Tr}(H v r^* P_{zw} W P_{zw}^* r v^*) = (r^* M r)(v^* H v)$. The coefficient of t^2 is negative, so $J(K_t) \rightarrow -\infty$ as $t \rightarrow \infty$. The corresponding controller $C_t = (I + D_t P_{zu})^{-1} D_t$ exists for all t except at most one value, because $\det(I + t vr^* P_{zu}) = 1 + t r^* P_{zu} v$. $\square$

C Cost of the stationary controller

Value at a stationary point. Let $K = D P_{zw}$ satisfy (20). Multiplying (20) on the right by D^* and using $K^* = P_{zw}^* D^*$ gives $(HK + N)WK^* = 0$; taking the trace,

$$\operatorname{Tr}(HKWK^*) = -\operatorname{Tr}(NWK^*). \quad (75)$$

The left side is real (H and KWK^* are Hermitian), so $\operatorname{Tr}(NWK^*)$ is real, and by (62) $\operatorname{Re} \operatorname{Tr}(N^* KW) = \operatorname{Re} \operatorname{Tr}(WK^* N) = \operatorname{Tr}(NWK^*)$. Substituting both into (18),

$$J(K) = J_0 + \frac{1}{2} \operatorname{Tr}(NWK^*). \quad (76)$$

All stationary points have the same cost. Let $K_1 = D_1 P_{zw}$ and $K_2 = D_2 P_{zw}$ both satisfy (20), and $\delta K = K_2 - K_1$. Subtracting the two instances of (20) gives $H \delta K W P_{zw}^* = 0$. By (19) at K_1 (whose first-order term vanishes, as in the proof of Proposition 2(a)), $J(K_2) - J(K_1) = \frac{1}{2} \operatorname{Tr}(H \delta K W \delta K^*) = \frac{1}{2} \operatorname{Tr}(H \delta K W P_{zw}^* (D_2 - D_1)^*) = 0$. This holds whatever the sign of H . $\square$

Proof of Proposition 3. It remains to evaluate (76) at

$$K_{\text{opt}} = D_{\text{opt}} P_{zw} = -H^\dagger N W P_{zw}^* M^\dagger P_{zw}. \quad (77)$$

Since $H^\dagger$ and $M^\dagger$ are Hermitian,

$$K_{\text{opt}}^* = -P_{zw}^* M^\dagger P_{zw} W N^* H^\dagger, \quad (78)$$

and

$$\operatorname{Tr}(NWK_{\text{opt}}^*) = -\operatorname{Tr} \left(N \left[W P_{zw}^* M^\dagger P_{zw} W \right] N^* H^\dagger \right). \quad (79)$$

By (67) and (29), the bracket is

$$W P_{zw}^* M^\dagger P_{zw} W = W^{1/2} Z^* (Z Z^*)^\dagger Z W^{1/2} = W^{1/2} Z^\dagger Z W^{1/2} = W^{1/2} \Pi_z W^{1/2}. \quad (80)$$

Using the cyclic property of the trace, $\operatorname{Tr}(NWK_{\text{opt}}^*) = -\operatorname{Tr}(N^* H^\dagger N W^{1/2} \Pi_z W^{1/2})$, and (76) gives (30). When $H \succeq 0$, $H^\dagger \succeq 0$; since $\Pi_z = \Pi_z^* = \Pi_z^2$, the trace equals $\|(H^\dagger)^{1/2} N W^{1/2} \Pi_z\|_F^2$. $\square$

Proof of Corollary 4. With $y_a = y_b$, $A = B = P_{yu}$ and $a = b = P_{yw}$, so $H = 2A^*A$, $N = 2A^*a$, $J_0 = \text{Tr}(aWa^*)$, and $H^\dagger = \frac{1}{2}(A^*A)^\dagger$. Since $A(A^*A)^\dagger A^* = AA^\dagger = \Pi_u$,

$$N^*H^\dagger N = 2a^*A(A^*A)^\dagger A^*a = 2a^*\Pi_u a. \quad (81)$$

So (30) becomes

$$J_{\text{opt}} = \text{Tr}(aWa^*) - \text{Tr}(W^{1/2}a^*\Pi_u aW^{1/2}\Pi_z). \quad (82)$$

Split $\text{Tr}(aWa^*) = \text{Tr}((I - \Pi_u)aWa^*) + \text{Tr}(W^{1/2}a^*\Pi_u aW^{1/2})$. Then

$$J_{\text{opt}} = \text{Tr}((I - \Pi_u)aWa^*) + \text{Tr}\left(W^{1/2}a^*\Pi_u aW^{1/2}(I - \Pi_z)\right). \quad (83)$$

Because Π_u and $I - \Pi_z$ are Hermitian and idempotent, with $E \doteq \Pi_u aW^{1/2}(I - \Pi_z)$ the second trace equals $\text{Tr}(EE^*) = \|E\|_F^2$, which is (31). $\square$

D Structure-constrained controllers: proofs

Proof of (32). Write $\epsilon = (I - CP_{zu})^{-1}$, so $D = \epsilon C$. The derivative of a matrix inverse gives $\delta\epsilon = \epsilon\delta C P_{zu} \epsilon$ to first order, since $\delta(I - CP_{zu}) = -\delta C P_{zu}$. Hence

$$\delta D = \delta\epsilon C + \epsilon\delta C = \epsilon\delta C(I + P_{zu}\epsilon C). \quad (84)$$

By the push-through identity $\epsilon C = CS_o$, and $I + P_{zu}CS_o = (S_o^{-1} + P_{zu}C)S_o = S_o$, which gives (32). $\square$

Proof of (33). With $\delta K = \epsilon\delta C S_o P_{zw}$, the first-order term of (19) is, by the cyclic property of the trace,

$$\text{Re Tr}((HK + N)WP_{zw}^*S_o^*\delta C^*\epsilon^*) = \text{Re Tr}(\epsilon^*(HK + N)WP_{zw}^*S_o^*\delta C^*). \quad (85)$$

For the relation to (6), (66) gives $\mathcal{G} = S_o \mathcal{P}(C)\epsilon = S_o[(HK + N)WP_{zw}^*]^*\epsilon = \Gamma^*$. $\square$

Proof of (38). Let $c(t)$ be the (real) impulse response of C , with $C(i\omega) = \int c(t)e^{-i\omega t}dt$, and let δc be an admissible perturbation supported in $\mathcal{S}$. Parseval's theorem, with the same transform convention for Γ and γ in (37), gives

$$\langle \Gamma, \delta C \rangle = \text{Re} \int_{-\infty}^{\infty} \text{Tr}(\gamma(t)\delta c(t)^T) dt = \int_{\mathcal{S}} \text{Tr}(\gamma(t)\delta c(t)^T) dt, \quad (86)$$

since γ and δc are real. This vanishes for every real δc supported in $\mathcal{S}$ if and only if $\gamma(t) = 0$ on $\mathcal{S}$ (almost everywhere, or at the point $t = 0$ when $\mathcal{S} = \{0\}$ and $\delta c(t) = \delta C\delta(t)$). For $\mathcal{S} = \{0\}$ the condition is $\text{Tr}(\gamma(0)\delta C^T) = 0$ for every real δC , i.e. $\gamma(0) = 0$; $\gamma(0)$ is real because $\Gamma(-\omega) = \overline{\Gamma(\omega)}$. $\square$

Proof of Proposition 5(a). For a stabilizing constant gain, the closed loop from w to $y = [Q^{1/2}x; R^{1/2}u]$ is $\dot{x} = A_c x + B_u w$, $u = Cx$. By Parseval's theorem, the total cost (5) computed in the frequency domain equals the time-domain value

$$\bar{J}(C) = \text{Tr}((Q + C^T R C) \Sigma_C), \quad (87)$$

for every stabilizing C . Two expressions for the same differentiable function of C have the same gradient. From the frequency side, for a real constant δC , $\delta \bar{J} = \langle \Gamma, \delta C \rangle = \text{Tr}(\gamma(0) \delta C^T)$, so the gradient is $\gamma(0)$. (Differentiation under the integral is justified because, for a stabilizing C , Γ decays like ω^{-2} uniformly in a neighbourhood of C .) From the time side, let $M \doteq Q + C^T R C$. Differentiating the first equation of (41), using $\delta A_c = B_u \delta C$,

$$A_c \delta \Sigma + \delta \Sigma A_c^T + B_u \delta C \Sigma_C + \Sigma_C \delta C^T B_u^T = 0. \quad (88)$$

Using the second equation of (41),

$$\begin{aligned} \text{Tr}(M \delta \Sigma) &= -\text{Tr}((A_c^T P_C + P_C A_c) \delta \Sigma) = -\text{Tr}(P_C (A_c \delta \Sigma + \delta \Sigma A_c^T)) \\ &= \text{Tr}(P_C (B_u \delta C \Sigma_C + \Sigma_C \delta C^T B_u^T)) = 2 \text{Tr}(\Sigma_C P_C B_u \delta C). \end{aligned} \quad (89)$$

Also $\text{Tr}(\delta M \Sigma_C) = 2 \text{Tr}(\Sigma_C C^T R \delta C)$. So

$$\delta \bar{J} = 2 \text{Tr}(\Sigma_C (P_C B_u + C^T R) \delta C) = \text{Tr}([2(B_u^T P_C + R C) \Sigma_C] \delta C^T), \quad (90)$$

and equating the two gradients gives (42). $\square$

Proof of Proposition 5(b). If $\Sigma_C \succ 0$, then by (42), $\gamma(0) = 0$ if and only if $RC + B_u^T P_C = 0$, i.e. $C = -R^{-1} B_u^T P_C$. Substituting this C into the second equation of (41),

$$\begin{aligned} 0 &= (A - B_u R^{-1} B_u^T P_C)^T P_C + P_C (A - B_u R^{-1} B_u^T P_C) + Q + P_C B_u R^{-1} B_u^T P_C \\ &= A^T P_C + P_C A - P_C B_u R^{-1} B_u^T P_C + Q, \end{aligned} \quad (91)$$

so P_C solves the algebraic Riccati equation, and $A - B_u R^{-1} B_u^T P_C = A_c$ is Hurwitz. The Riccati equation has at most one stabilizing solution, so $P_C = P$ and $C = -R^{-1} B_u^T P = -K_{\text{LQR}}$. Conversely, at $C = -K_{\text{LQR}}$, $P_C = P$ and $RC + B_u^T P = 0$, so $\gamma(0) = 0$. $\square$

E Causal optimum: proofs

Proof of (44). Put $K = K_0 + \delta K$ with $\delta K = T_2 Q T_3$ in (19). The first-order term is

$$\text{Re Tr}((H K_0 + N) W T_3^* Q^* T_2^*) = \text{Re Tr}(T_2^* (H K_0 + N) W T_3^* Q^*) = \text{Re Tr}(E Q^*), \quad (92)$$

and the second-order term is $\frac{1}{2} \text{Tr}(H T_2 Q T_3 W T_3^* Q^* T_2^*) = \frac{1}{2} \text{Tr}(T_2^* H T_2 Q T_3 W T_3^* Q^*)$, by the cyclic property of the trace. $\square$

Proof of Theorem 6. Let $Y \doteq \Lambda Q \Gamma$. By (47), $\text{Tr}(\tilde{H} Q \tilde{M} Q^*) = \text{Tr}(\Lambda^* \Lambda Q \Gamma \Gamma^* Q^*) = \|Y\|_F^2$ at each frequency. By (48), $E = \Lambda^* F \Gamma^*$, so

$$\text{Re Tr}(EQ^*) = \text{Re Tr}(\Lambda^* F \Gamma^* Q^*) = \text{Re Tr}(F(\Lambda Q \Gamma)^*) = \text{Re Tr}(FY^*). \quad (93)$$

Integrating (44) over frequency and completing the square,

$$\bar{J} = \bar{J}(K_0) + \frac{1}{2}\|Y + F\|^2 - \frac{1}{2}\|F\|^2. \quad (94)$$

Because Λ , Γ and their inverses are stable and causal, Q is stable and causal if and only if Y is. Stable, causal, square-integrable functions are exactly those with $Y = [Y]_+$, and they are orthogonal to anticausal ones. Hence

$$\|Y + F\|^2 = \|Y + [F]_+\|^2 + \|[F]_-\|^2, \quad (95)$$

which is minimized over causal Y by $Y = -[F]_+$, giving (49), with $\bar{J} = \bar{J}(K_0) - \frac{1}{2}\|F\|^2 + \frac{1}{2}\|[F]_-\|^2$. Without the causality constraint, (94) is minimized frequency by frequency by $Y = -F$, i.e. $Q_{\text{pw}} = -\Lambda^{-1} F \Gamma^{-1}$, with cost $\bar{J}_{\text{pw}} = \bar{J}(K_0) - \frac{1}{2}\|F\|^2$. This gives (50). For a stable plant, $K_0 = 0$, $T_2 = I$, $T_3 = P_{zw}$, and $Q_{\text{pw}} = -\Lambda^{-1} \Lambda^{-*} N W P_{zw}^* \Gamma^{-*} \Gamma^{-1} = -H^{-1} N W P_{zw}^* M^{-1}$. $\square$

F Greedy selection: proofs

Proof of Proposition 7. With real-rational data, Λ , Γ and their inverses are rational, stable and causal, so Λ^{-*} and Γ^{-*} (the para-Hermitian conjugates, $X^*(s) = X(-\bar{s})^*$) have all their poles in the open right half-plane. Hence every left-half-plane pole of $F = \Lambda^{-*} E \Gamma^{-*}$ is a pole of E . By partial fractions, the causal part $[F]_+$ of a rational F with no poles on the imaginary axis is the sum of the partial-fraction terms at its left-half-plane poles (plus, by our convention, half of any constant term). So $[F]_+$ is rational with poles among the left-half-plane poles of E . A Takenaka-Malmquist basis built from these poles (with multiplicity) spans all strictly proper stable rational functions with those poles, so $Y_c = -[F]_+$ is a finite combination of its elements. $\square$

Proof of (58). With new coefficients β for the candidate columns, the cost is $c + b^T \theta + b_{\psi}^T \beta + \frac{1}{2}[\theta; \beta]^T \begin{bmatrix} \mathbf{H} & \mathbf{H}_{\theta\psi} \\ \mathbf{H}_{\psi\theta} & \mathbf{H}_{\psi\psi} \end{bmatrix} [\theta; \beta]$. Minimizing over θ for fixed β gives $\theta = \theta^* - \mathbf{H}^{-1} \mathbf{H}_{\theta\psi} \beta$ with $\theta^* = -\mathbf{H}^{-1} b$, and the remaining function of β is $\bar{J}(\theta^*) + \rho^T \beta + \frac{1}{2} \beta^T S \beta$. Its minimum over β is $\bar{J}(\theta^*) - \frac{1}{2} \rho^T S^{-1} \rho$. $\square$

G Sensor-noise sensitivity: proof

Proof of Proposition 8. With $Q = D$ and a stable plant, $K_0 = 0$, $T_2 = I$, $T_3 = [P_{zw}, I]$ and $W = \text{blkdiag}(W_w, V)$. The noise columns of $P_{y_{aw}}$ and $P_{y_{bw}}$ are zero, so $N W T_3^* = N_w W_w P_{zw}^*$, i.e. E does not depend on V , while $\tilde{M} = T_3 W T_3^* = P_{zw} W_w P_{zw}^* + V$. By

(44), for fixed Q the only V -dependent term is $\frac{1}{2}\text{Tr}(\tilde{H}QVQ^*) = \frac{1}{2}\text{Tr}(Q^*\tilde{H}QV)$, which is linear in V ; this gives (59). Since $\tilde{H} \succeq 0$, $Q^*\tilde{H}Q \succeq 0$, so $\bar{J}(Q; V)$ is non-decreasing in V (in the positive-semidefinite order). $\bar{J}_c(V)$ is a pointwise minimum over Q of functions that are affine and non-decreasing in V , hence concave and non-decreasing. Where the minimizer Q_c is unique and depends smoothly on V , the envelope theorem (the first-order term in Q vanishes at Q_c) gives $\partial \bar{J}_c / \partial V = \partial \bar{J} / \partial V|_{Q=Q_c}$, which is (60). The same argument applies to the pointwise problem at each frequency. $\square$

Proof of Corollary 9. By (59), with $Q = Q_c$ fixed,

$$\bar{J}(Q_c; V') = \bar{J}(Q_c; V) + \frac{1}{2\pi} \int \frac{1}{2}\text{Tr}(Q_c^*\tilde{H}Q_c(V' - V)) d\omega$$

exactly, because the cost is affine in V for fixed Q . Q_c is causal and stable, hence admissible for V' (the parameterization $Q = D$ does not depend on V), so $\bar{J}_c(V') \leq \bar{J}(Q_c; V')$. Using $\bar{J}(Q_c; V) = \bar{J}_c(V)$ and rearranging gives (61). $\square$

Proof of Proposition 10. Write $Q = \Sigma^{-1}$ and $W = BB^T$. Multiplying the Riccati inequality $A\Sigma + \Sigma A^T + W - \Sigma C^T \text{diag}(p) C \Sigma \preceq 0$ on both sides by Q gives the equivalent information form

$$\mathcal{F}(Q, p) \doteq QA + A^T Q + QWQ - C^T \text{diag}(p) C \preceq 0. \quad (96)$$

$Q \mapsto QWQ$ is matrix convex for $W \succeq 0$ and the remaining terms are linear in (Q, p) , so $\mathcal{F}$ is matrix convex and $\mathcal{C} \doteq \{(Q, p) : Q \succ 0, p \geq 0, \mathcal{F}(Q, p) \preceq 0\}$ is convex. For fixed p , every $\Sigma \succ 0$ satisfying the Riccati inequality satisfies $\Sigma \succeq \Sigma_*(p)$ (the stabilizing solution is the minimal such solution), with equality attained, so

$$\text{Tr}(G \Sigma_*(p)) = \min\{\text{Tr}(GQ^{-1}) : (Q, p) \in \mathcal{C}\}. \quad (97)$$

$Q \mapsto \text{Tr}(GQ^{-1})$ is convex on $Q \succ 0$ for $G \succeq 0$, so the right-hand side of (97) is the partial minimization of a jointly convex function over a convex set, hence convex in p . The statement for (99) follows because $\text{Tr}(PBB^T)$ does not depend on p and $\Sigma_e(\tau) = e^{A\tau}\Sigma_*e^{A^T\tau} + \int_0^\tau e^{As}WW^Te^{A^Ts}ds$ contributes $\text{Tr}(G\Sigma_*(p))$ plus a constant. $\square$

H Exact reference for measurement latency

The examples of Section 7 are checked against an exact optimum that we derive here for completeness; it is the standard separation argument, specialized to a pure measurement delay. Let $\dot{x} = Ax + B_u u + B_w \xi$ with ξ white of unit intensity (coloured disturbances are included by augmenting their shaping filters to x), A stable, cost $\mathbb{E}[x^T Q_x x + u^T R u]$ with $R \succ 0$, and measurements $z(t) = Cx(t - \tau) + v(t)$ with v white of intensity $V \succ 0$, independent of ξ . Let P solve the control Riccati equation and $K = R^{-1}B_u^T P$.

Cost identity. For any input u that is a causal function of the measurements, Itô's formula for $x^T P x$ and stationarity give $\mathbb{E}[x^T Q_x x + u^T R u] = \text{Tr}(PB_w B_w^T) + \mathbb{E}[(u + Kx)^T R (u + Kx)]$.

Optimal input. The information available at time t is $\mathcal{Z}_t = \{z(s) : s \leq t\}$, which determines x only up to time $t - \tau$. Write $\hat{x}(t) = \mathbb{E}[x(t) \mid \mathcal{Z}_t]$. Because the input is known to the controller, the error $e = x - \hat{x}$ does not depend on the input (linear–Gaussian model), so $\mathbb{E}[(u + Kx)^T R(u + Kx)] = \mathbb{E}[(u + K\hat{x})^T R(u + K\hat{x})] + \text{Tr}(K^T R K \Sigma_e)$ with $\Sigma_e = \mathbb{E}[ee^T]$ independent of u . The minimum is attained by $u = -K\hat{x}$.

Estimate and error. From $x(t) = e^{At}x(t-\tau) + \int_0^\tau e^{Ar}(B_u u(t-r) + B_w \xi(t-r)) dr$, and the independence of ξ on $(t - \tau, t]$ from $\mathcal{Z}_t$,

$$\hat{x}(t) = e^{A\tau} \hat{x}_f(t) + \int_0^\tau e^{Ar} B_u u(t-r) dr, \quad \Sigma_e(\tau) = e^{A\tau} S e^{A^T \tau} + \int_0^\tau e^{Ar} B_w B_w^T e^{A^T r} dr, \quad (98)$$

where $\hat{x}_f(t) = \mathbb{E}[x(t - \tau) \mid \mathcal{Z}_t]$ is the steady-state Kalman filter for the delayed plant (driven by $u(t - \tau)$ and $z(t)$) and S its error covariance, the stabilizing solution of the filter Riccati equation (it does not depend on τ). Hence the exact optimal cost is

$$\bar{J}_c(\tau) = \text{Tr}(P B_w B_w^T) + \text{Tr}(K^T R K \Sigma_e(\tau)), \quad (99)$$

non-decreasing in τ , because $\Sigma_e(\tau)$ is the error covariance of estimating $x(t)$ from measurements of the state up to $t - \tau$, and less information cannot reduce it. In the frequency domain, with $\Phi_f = (sI - A + LC)^{-1}$ and $L = SC^T V^{-1}$, (98) gives $\hat{X} = e^{A\tau} \Phi_f (B_u e^{-s\tau} U + LZ) + (sI - A)^{-1} (I - e^{-s\tau} e^{A\tau}) B_u U$, and $U = -K \hat{X}$ yields the controller frequency response used in Figures 8 and 9. The integral in (98) is computed with Van Loan’s matrix exponential. When the plant is only known through frequency responses, or the disturbance spectrum is not rational, this construction is unavailable while the frequency-domain method of Section 4 applies unchanged; here the state-space model is used only to check it.

I PID designs on the Wood–Berry column: comparison

Figure 14 compares the designs of Section 7.4. It is produced by the script `demo_pid_woodberry.py` in `src/examples/`, which also prints every number quoted there; the plant, the design parameters and the published gains are those of [4, §8].

The top row shows the three closed-loop transfer functions that carry the constraints in their formulation, with the imposed limits in red. The design obtained from the spectral cost alone is drawn dashed because its closed loop has two right-half-plane poles: on the imaginary axis it looks unexceptional, which is the point made in Section 3 — the cost is an integral over that axis and does not see the instability. The bottom row shows the cost density and the closed-loop step responses, obtained from the same frequency responses by inverse transform, so that no separate time-domain model is involved. The unstable design is omitted from the time-domain panels.

The design driven by an integrating disturbance tracks the published one closely in every panel, which is the sense in which the two formulations agree once the disturbance spectrum is chosen to weight the frequencies their objective is about. The design driven by the first-order disturbance spectrum settles more slowly in the second loop: its cost

has no term that asks for tracking, and the peak constraints are what stop it from removing the integral action altogether.

J Problem statements for the worked examples

Each example is stated here in full, so that it can be read without the sources it is taken from, and each ends with the runtime of the script that produces the numbers in the main text. Times are wall-clock, single process, on a laptop, measured in one sequential run of all the scripts. They are dominated by the reference computations and the searches rather than by the frequency-domain design: the four demonstrations of Section 6 together take 71 s, and none of the designs below takes more than a few minutes on its own.

J.1 Stylized two-area power system

Plant. Ten classical machines, five per area, linearized swing equations with lossless DC power flow:

$$\begin{aligned}\dot{\theta}_i &= \omega_i, \\ M_i \dot{\omega}_i &= -\sum_j k_{ij}(\theta_i - \theta_j) - D\omega_i - \beta \sum_j k_{ij}(\omega_i - \omega_j) + p_{m,i} + u_i - \ell_i, \\ T_g \dot{p}_{m,i} &= -p_{m,i} - \omega_i/(R_g \omega_s),\end{aligned}$$

with $M_i = 2H_i/\omega_s$, H_i between 3 and 6.5 s, $\omega_s = 2\pi 60$, $R_g = 0.05$, $T_g = 2$ s. Intra-area line susceptances are 0.8–1.6 pu and the two tie lines are 0.2 and 0.25 pu, which puts the inter-area mode at 0.43 Hz. The term in $\beta = 0.006$ is proportional to the network Laplacian and stands in for local damping (damper windings, local stabilizers): it damps local modes more than the inter-area mode. Angles enter only through differences, so machine 10 is the reference and the plant has 29 states. The full parameter list is in `src/examples/interarea_model.py`.

Disturbance. A load fluctuation at every bus, $\dot{\ell}_i = -a\ell_i + \sqrt{2a}\sigma \xi_i$, with $a = 1$ rad/s and $\sigma = 0.01$ pu.

Actuators and sensors. Two fast power injections at buses 2 and 9. A candidate PMU at each bus measures the machine frequency ω_i with white noise of intensity $V_0 = 10^{-4}(\text{rad/s})^2\text{s}$ (about 11 mHz rms at 50 frames/s), delivered with latency τ : the controller at time t may use measurements up to $t - \tau$.

Objective. The sum of squared tie-line power deviations (units 0.01 pu), machine frequencies relative to the centre of inertia (units 10 mHz, averaged over machines), the centre-of-inertia frequency (units 10 mHz, weight 0.1), and actuator powers (units 0.02 pu).

Questions. (i) the cost of latency and whether the frequency-domain tuner attains it; (ii) which k PMUs to install, with all $\binom{10}{k}$ subsets enumerable; (iii) how to allocate accuracy when precision has a price.

Reference solution. Exact delayed-measurement LQG, (99), computed from Riccati equations on the model augmented with the load filters (39 states). Runtime of

`demo.interarea.py`: 917 s, almost all of it in the four greedy tuning runs (about 210–300 s each); the exhaustive search over all 1023 PMU subsets takes 4 s.

J.2 IEEE 39-bus system, classical-machine model

Data. Network, loads, AC power-flow solution, machine inertias and transient reactances (GENROU $M = 2H$, x'_d , r_a) and governor parameters (TGOV1: R , T_1 , T_2 , T_3) from the IEEE 39-bus case distributed with ANDES 2.0 (`ieee39_full.xlsx`, power flow solved with ANDES). The extracted data are in `src/examples/data/ieee39.json`; `make_ieee39_data.py` regenerates them.

Model and simplifications. Classical machines, constant EMF behind $r_a + jx'_d$, with the dataset’s governors and no added damping (the dataset has $D = 0$); exciters and power system stabilizers omitted; constant-power loads; AC network equations linearized at the power-flow solution and the algebraic variables eliminated exactly. Our linearization reproduces the ANDES power flow to a residual of 1.8×10^{-5} , and the machine electrical powers to machine precision. Angles are relative to machine 10, giving 79 plant states (98 with the load filters). Because exciters are omitted, modal damping here is higher than in the full benchmark: the results are for this simplified model.

Disturbance. Ornstein–Uhlenbeck load fluctuations at the 19 load buses, corner 1 rad/s, standard deviation 1% of each load.

Actuators and sensors. Two fast power injections acting on the swing equations of the machines at buses 33 and 38 — the pair with the lowest full-information cost among the 36 generator pairs that exclude the external-system equivalent at bus 39. A candidate PMU at each of the 39 buses measures bus frequency through a first-order filter with $T_f = 0.05$ s (standing in for the PMU’s frequency estimation), with latency τ and white noise of intensity 10^{-6} (rad/s)²s, about 1.1 mHz rms at 50 frames/s against open-loop bus-frequency deviations of about 4.3 mHz rms.

Objective. The sum of the swing-driven part of the flows on the five tie lines between the dataset’s two areas (units 0.01 pu), machine frequencies relative to the centre of inertia (units 1 mHz), the centre-of-inertia frequency (weight 0.1), and actuator powers (units 0.05 pu). The instantaneous redistribution of a load change over the network is excluded, because no controller acting on the machines can affect it.

Questions. As in the two-area case, but with 39 candidate sensors, so that exhaustive search is possible only for small sets.

Reference solution. As above. Runtime of `demo_ieee39.py`: 401 s with the exhaustive search read from `data/ieee39_exhaustive.json`, and about 57 minutes when it is recomputed — the 82 251 sets of four PMUs alone take 3026 s. Forward greedy over all 39 sizes takes 37 s, and each of the two tuning runs about 124 s. Deleting the cache file recomputes everything.

J.3 Vibration isolation with two dissimilar sensors

Plant. A platform on a passive isolator, $\ddot{x} = -\omega_0^2(x - x_g) - 2\zeta_0\omega_0(\dot{x} - \dot{x}_g) + u$, with $\omega_0 = 2\pi$ rad/s and $\zeta_0 = 0.05$; x is the inertial platform displacement in micrometres and

u a force. The numbers are illustrative and do not model a specific instrument.

Disturbance. Ground motion x_g with a microseism-like spectrum: white noise through a second-order filter at 0.15 Hz with damping 0.3.

Sensors. A relative sensor $d = x - x_g + n_d$ with white noise ($V_d = 10^{-4}$), which sees the ground motion as if it were noise; and an inertial sensor $s = x + n_s$ whose noise spectrum is $V_s(\omega) = v_s((\omega^2 + \omega_c^2)/(\omega^2 + \omega_e^2))^2$ with $v_s = 10^{-3}$, $\omega_c = 2\pi \cdot 0.5$ and $\omega_e = 2\pi \cdot 0.01$ rad/s: flat above the corner, rising as f^{-4} in power below it, capped below ω_e .

Objective. $\mathbb{E}[x^2] + 10^{-4} \mathbb{E}[u^2]$.

Questions. What the optimal controller does with two sensors that are each poor in a different band; how much each sensor's accuracy is worth and at which frequencies; and what a candidate upgrade is guaranteed to buy.

Reference solution. LQG on the plant augmented with the ground and noise shaping filters. The inertial noise has a white part correlated with the shaping-filter input, so the filter Riccati equation carries the cross-covariance term. Runtime of `demo_fusion.py`: 88s.

J.4 MIMO PID on the Wood–Berry column

Plant. The Wood–Berry binary distillation column, with time in minutes,

$$P(s) = \begin{bmatrix} \frac{12.8 e^{-s}}{16.7s + 1} & \frac{-18.9 e^{-3s}}{21.0s + 1} \\ \frac{6.6 e^{-7s}}{10.9s + 1} & \frac{-19.4 e^{-3s}}{14.2s + 1} \end{bmatrix},$$

controlled by $C(s) = K_P + K_I/s + K_D s/(\tau s + 1)$ with $\tau = 0.3$ in negative feedback. The delays are used exactly; no rational approximation and no state-space realization are formed.

The LMI design problem [4, §8]. Minimize $\|(P(0)K_I)^{-1}\|$ — the low-frequency sensitivity, and hence a measure of tracking — subject to $\|S\|_\infty \leq 1.4$, $\|T\|_\infty \leq 1.4$ and $\|Q\|_\infty \leq 3/\sigma_{\min}(P(0)) = 0.738$, where $S = (I + PC)^{-1}$, $T = I - S$ and $Q = CS$. The semi-infinite constraints are sampled on a frequency grid, and the non-convex problem is solved by a convex restriction, iterated from a stabilizing initial design.

Our design problem. Minimize a spectral cost $\mathbb{E}|y|^2 + \rho \mathbb{E}|u|^2$ with $\rho = 0.05$ for an output disturbance, over the same PID structure, using the stationarity condition (38) with the admissible lag set $\{1, 1/s, s/(\tau s + 1)\}$. Two disturbance spectra are used: a first-order spectrum with corner 0.05 rad/min, and an integrating spectrum $W(\omega) = (\omega^2 + a^2)^{-1}$, which weights the frequencies that the LMI objective is about. The peak constraints are imposed as penalties on a sampled band, and stability is enforced by a Nyquist test: the number of encirclements of the origin by $\det(I + PC)$ along a contour indented to the right of the integrator poles, which must be zero.

Questions. Whether this framework can express and solve a published fixed-structure design problem with delays; what it gives when the objective is a spectral

cost rather than a peak-constrained tracking measure; and whether choosing the disturbance spectrum recovers the published design.

Reference solution. The published gains [4, §8], reproduced here to check the plant, the conventions and the metrics. Runtime of `demo_pid_woodberry.py`: 475s, dominated by the derivative-free stages that carry the peak penalties and the stability test; the gradient-based design itself takes seconds.

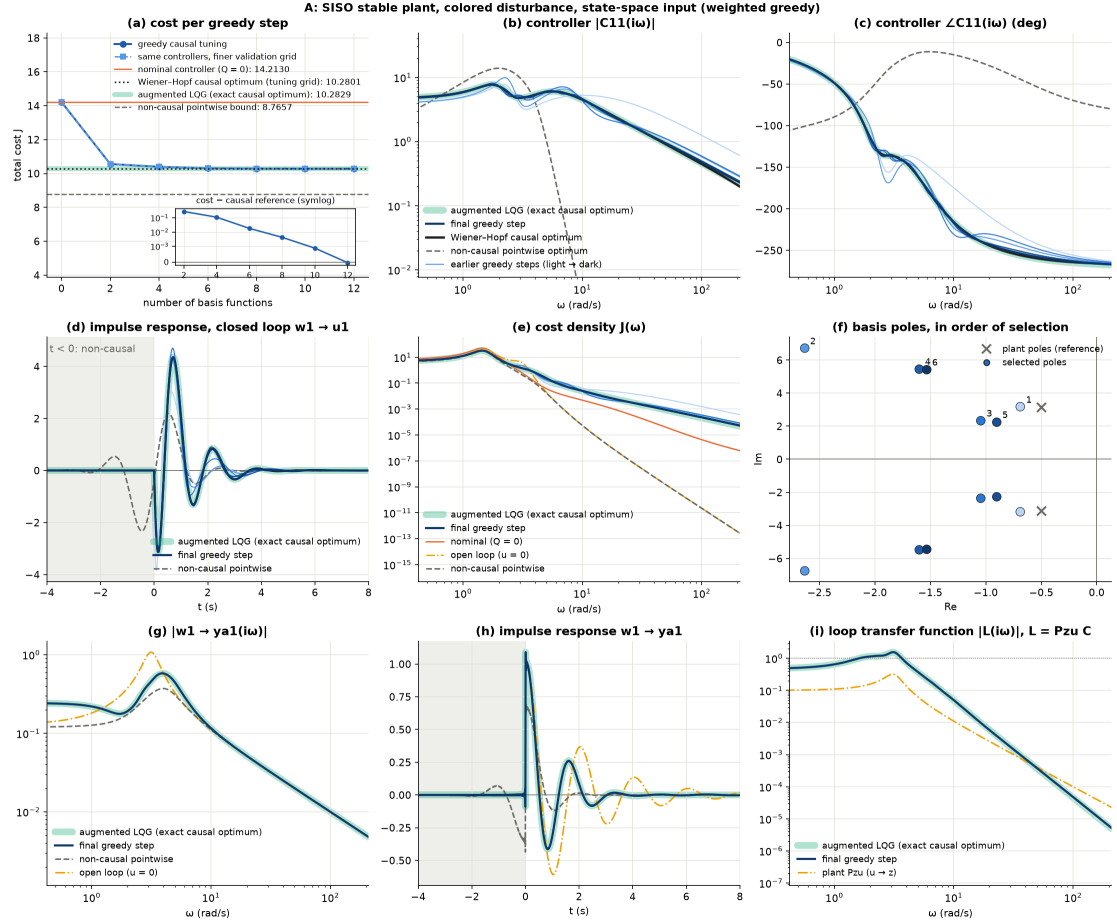


Figure 6: Case A of Table 2: output of `plot_autotune`. Greedy steps are drawn from light to dark blue, with the final step in dark navy; the wide translucent green band is the exact causal optimum (augmented LQG); the dashed curves are the non-causal pointwise optimum, which is not realizable (its impulse responses in panels (d) and (h) are non-zero for $t < 0$); dash-dot curves are the uncontrolled plant.

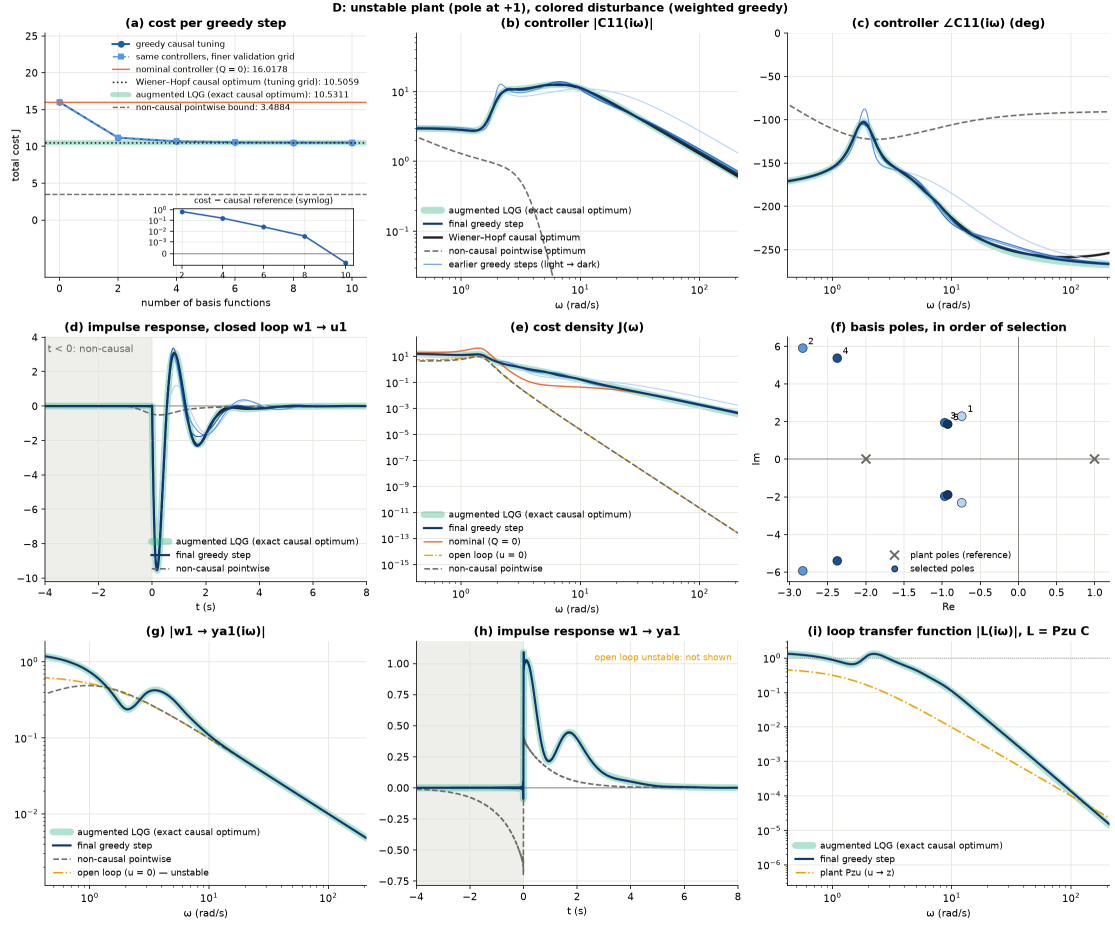


Figure 7: Case D of Table 2: unstable plant (pole at +1). The open-loop impulse response is omitted in panel (h) because the uncontrolled plant is unstable.

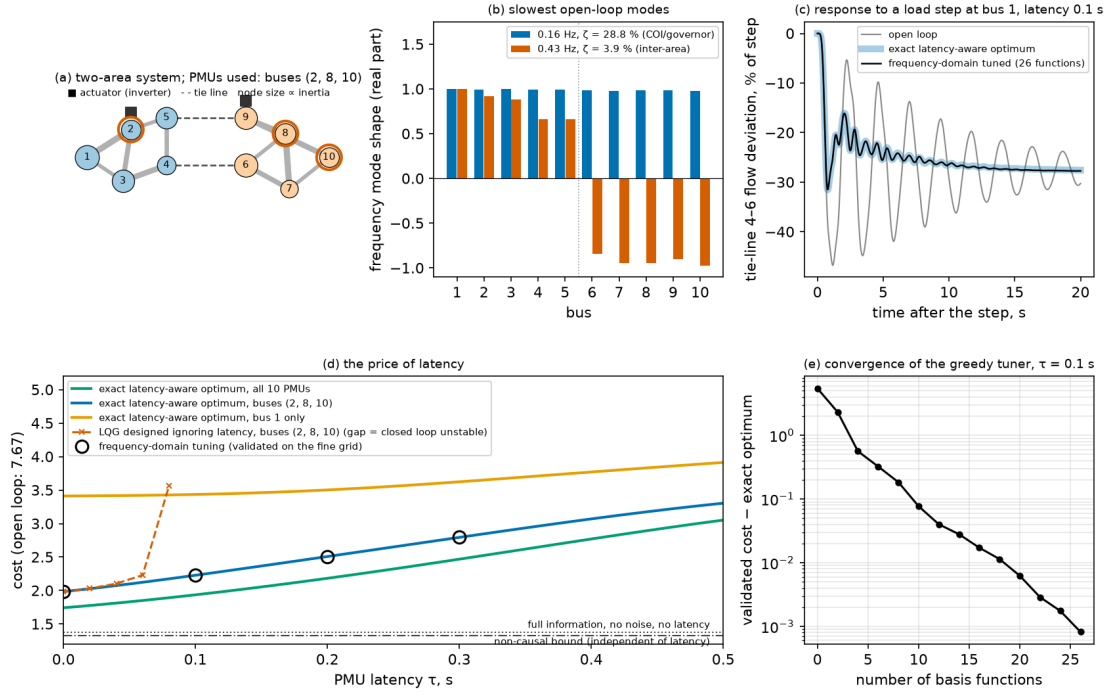


Figure 8: Stylized two-area system. (a) Network, actuators and the PMU set used. (b) Frequency mode shapes of the two slowest modes. (c) Tie-line response to a load step at bus 1, latency 0.1 s. The LQG controller designed ignoring the latency is unstable at this latency and is not shown. (d) Exact optimal cost against latency for three PMU sets, the latency-ignoring LQG design (missing where unstable), and the frequency-domain tuned controllers. (e) Convergence of the greedy tuner.

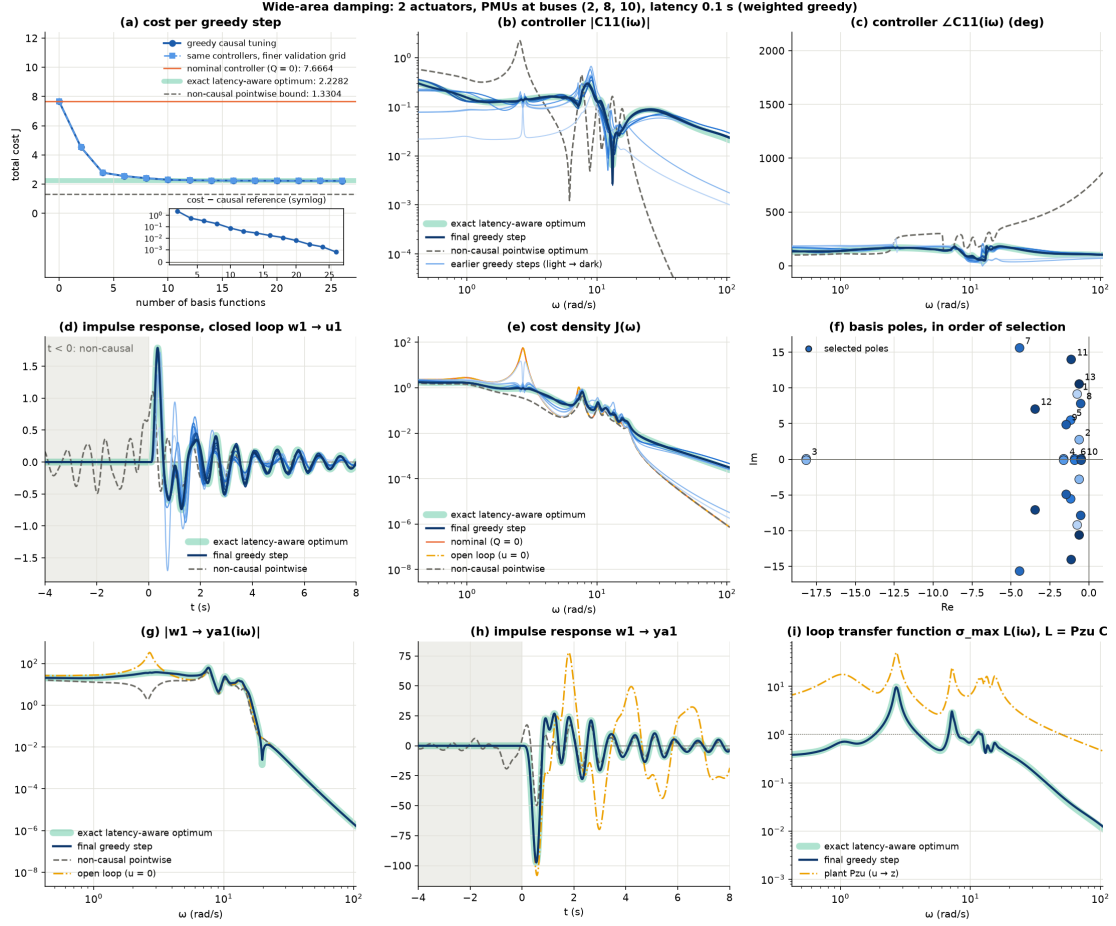


Figure 9: Two-area system, PMUs at buses 2, 8, 10, latency 0.1s: the `plot_autotune` report (Section 6). The exact latency-aware optimum (Appendix H) is drawn as the wide band.

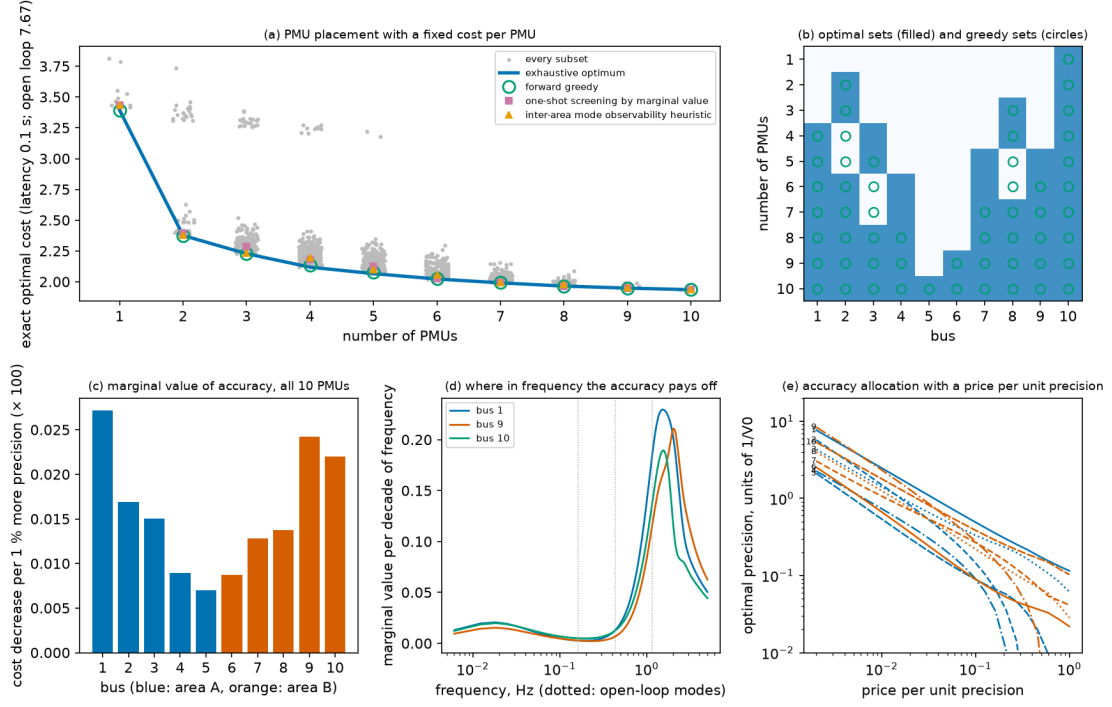


Figure 10: Two-area system, latency 0.1 s. (a) Exact optimal cost for every PMU subset (grey), the exhaustive optimum, greedy selection and two screening rules. (b) Optimal sets (filled) and greedy sets (circles). (c) Proposition 8 marginal value of each PMU's accuracy at the all-PMU design. (d) The same, resolved in frequency, per decade (area under the curve on the logarithmic axis is proportional to the contribution). (e) Optimal precision of each PMU against a linear price per unit precision.

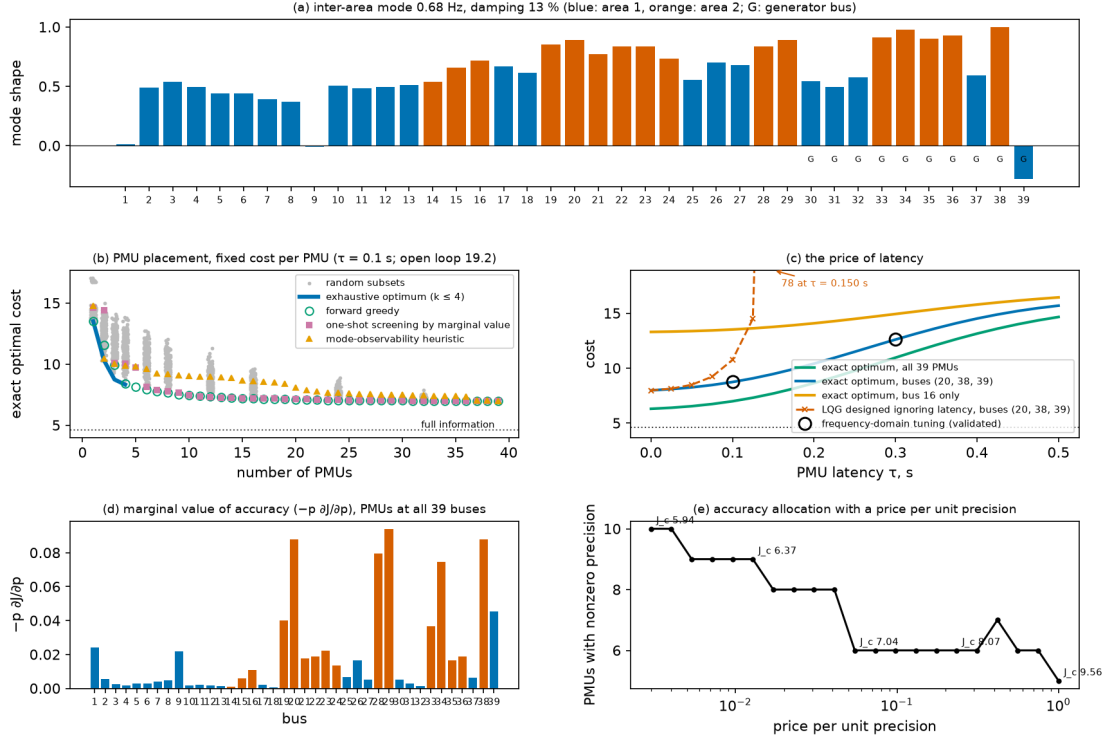


Figure 11: IEEE 39-bus model. (a) Bus-frequency shape of the 0.68 Hz inter-area mode. (b) Exact optimal cost against the number of PMUs: random sets (grey), exhaustive optimum ($k \leq 4$), greedy selection and two screening rules. (c) Exact optimal cost against latency for three PMU sets, the LQG design that ignores latency (missing where unstable), and the frequency-domain tuned controllers. (d) Proposition 8 marginal value of accuracy at the all-PMU design. (e) Number of PMUs with nonzero optimal precision against a linear price per unit precision.

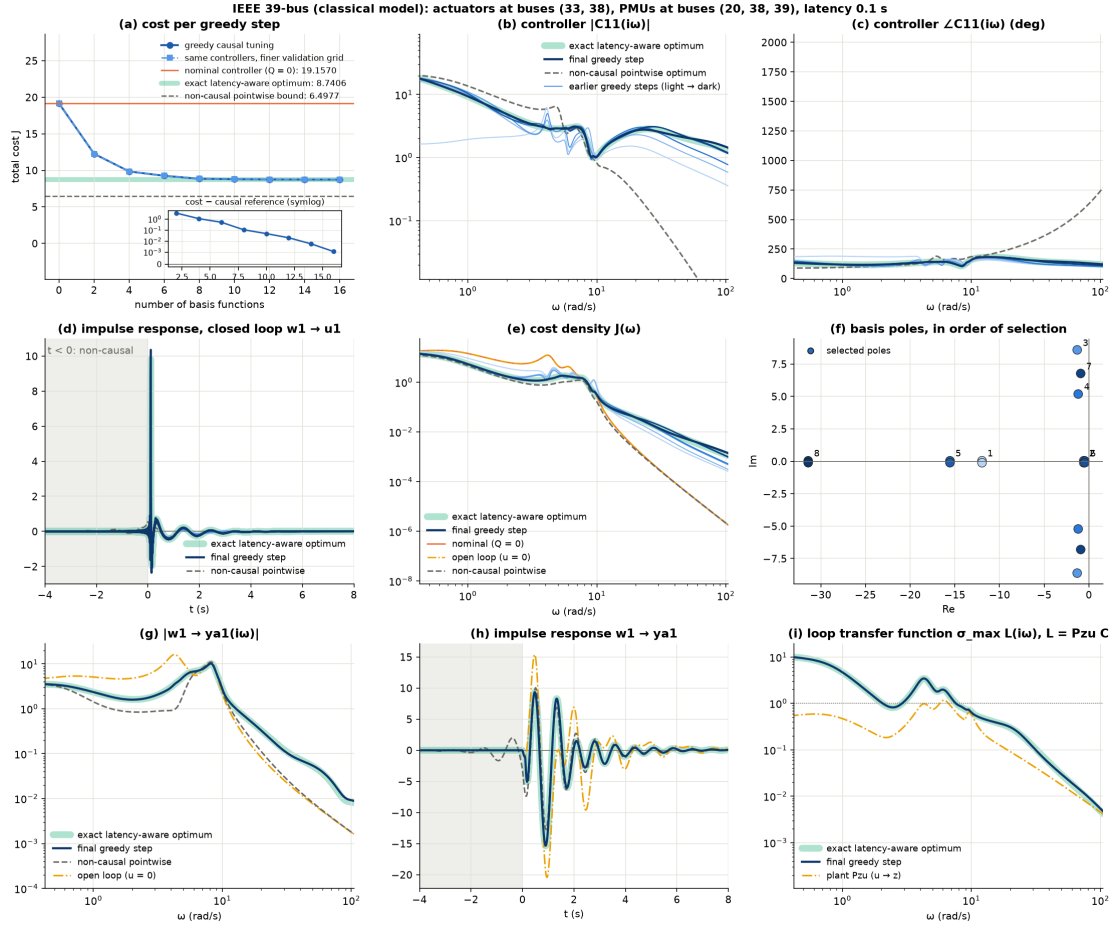


Figure 12: IEEE 39-bus model, PMUs at buses 20, 38, 39, latency 0.1s: the plot_autotune report. The exact latency-aware optimum is the wide band.

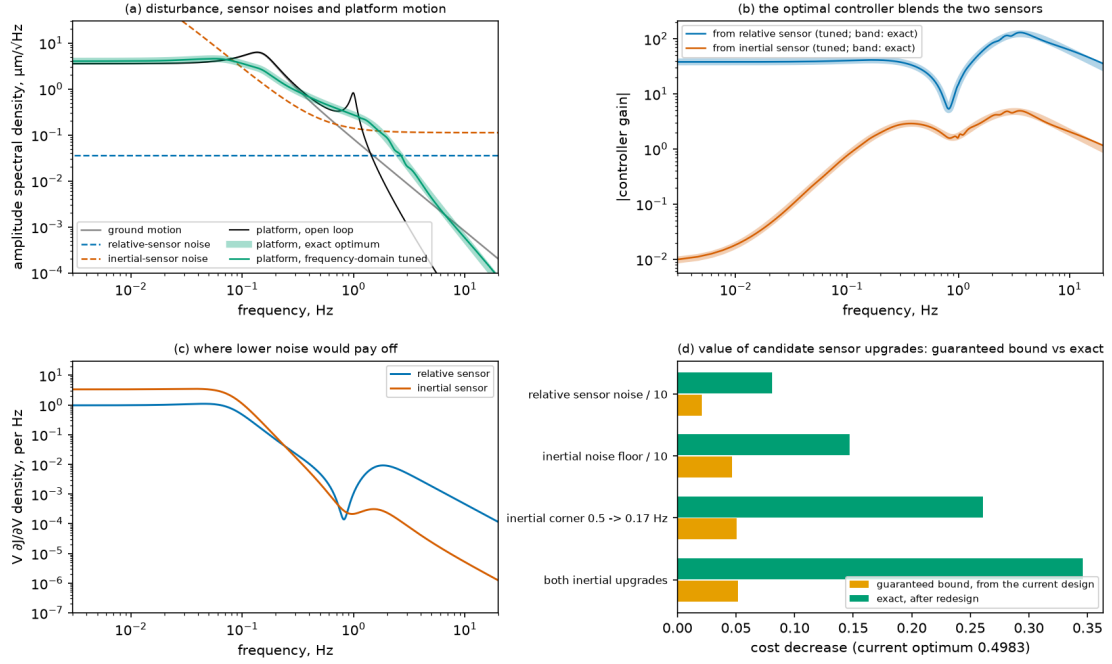


Figure 13: Sensor fusion (Section 7.3). (a) Ground-motion and sensor-noise spectra, and platform motion in open loop, at the exact optimum (wide band) and with the frequency-domain tuned controller. (b) Controller gain from each sensor: tuned (line) and exact (band). (c) Proposition 8 marginal value density of each sensor's noise. (d) Candidate upgrades: guaranteed decrease (61) from the current design, and exact decrease after redesign.

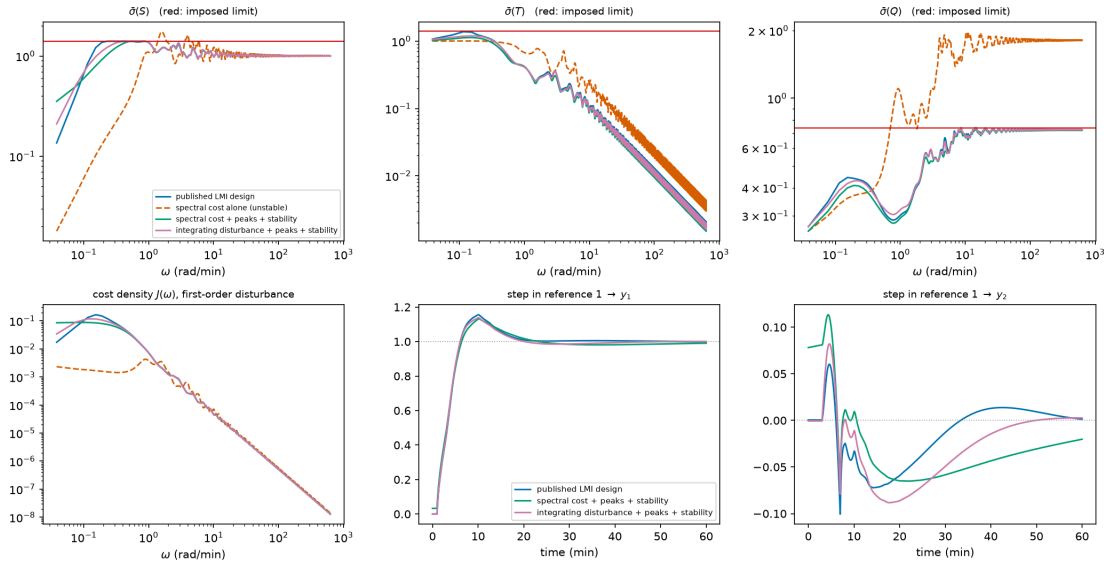


Figure 14: Wood–Berry column, MIMO PID (Section 7.4). Top: $\bar{\sigma}(S)$, $\bar{\sigma}(T)$ and $\bar{\sigma}(Q)$ against frequency, with the limits of [4] in red; the dashed curve is the design that minimizes the spectral cost alone, whose closed loop is unstable although nothing on the imaginary axis says so. Bottom: cost density, and the responses of both outputs to a unit step in the first reference, computed from the same frequency responses.